Phys7502 Class Today

- 3/26/2026
- No Office Hours Tomorrow 11am

- New HW Released Tonight

- Reading Next time: Read into thru Griff Ch 9.1 (Weak interactions) → Don't go further into Ch 9 for now)

- READ MANN ch 7/8 Accelerators and Detectors

Relative Minus in Rules

Feynman Rules for QED

2. External Lines: contribute factors to $\mathcal{M}$ as follows:

5. Conservation of energy and momentum: for each vertex write a factor

$$(2\pi)^4 \delta^4(k_1 + k_2 + k_3)$$

enforcing overall energy and momentum conservation at that vertex. (here each k represents a 4 momentum; incoming four momenta are positive and outgoing are negative).

6. Integrate over all internal momenta: for each internal momentum q write a factor of $(2\pi)^{-4} d^4q$ and integrate.

7. Cancelling the remaining δ-function (expressing overall energy and momentum conservation) leaves you with $-i\mathcal{M}$.

8. Anti-symmetrization: include a relative minus sign between diagrams differing only by the exchange of two incoming (or outgoing) electrons (or positrons) or of and incoming electron with an outgoing positron (or vice versa) [see next slide].

"Relative" Minus Sign in Dirac Feyman Rules

- Stack-exchange:

diagrams, involving fermions. A simple example is **Moller** scattering $e^- e^- \to e^- e^-$. This process can happen by scattering or annihilation.

(u-chan or t-chan)

We have $|i\rangle = |e^- e^-\rangle = c^\dagger d^\dagger |0\rangle$ and $\langle f| = \langle 0|dc$

these are creation op's like $a^\dagger$



The corresponding amplitudes are therefore (again focusing only o the sign relevant parts) (this part slightly mod'd by JF)

$$\langle f|S^a|i\rangle \propto \langle 0|cd \quad \{ \qquad \} c^\dagger d^\dagger |0\rangle$$

and

$$\langle f|S^b|i\rangle \propto \langle 0|dc \quad \{ \qquad \} c^\dagger d^\dagger |0\rangle$$

for Dirac fermions a, $a^\dagger$ "anticommute" instead of commute

Also explained there: why do we want all the operators in the same order?
Ans: It's needed by the way the calc framework is set up to factorize "outer" bracketing
Need Something like at end (factor out:

$$\{a_i, a^\dagger_j\} \propto \delta_{ij}$$

$$\mathcal{M}_{t+t} = \langle \mathcal{M}_1 \rangle + \langle \mathcal{M}_2 \rangle$$

$$\mathcal{M}_{t+t} = \langle 0|cd \left(\mathcal{M}_a - \mathcal{M}_b \right) c^\dagger d^\dagger |0\rangle$$

QFT Explains Pauli-Exclusion?

- Griffiths says this was one of the great successes of QFT

- I can't find too much to support this statement

- It's clear it's due to the Anti commutation relations of the creation/annihilation operators :

- Because it implies that
- **$aa = 0$ or $a^+a^+ = 0$!!!**

- Note anti-commutation of gamma's not directlu related !!!

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$$

$$\left\{\hat{c}_i^\dagger, \hat{c}_j^\dagger\right\} = 0,$$

Why Fermions/Exlc Principle Required

- Lancaster Blundell : start positing from the start

Whether you first put one particle in state p_1 and then another in state p_2, or do the same thing in the reverse order, you must end up with the same final state: $|11\rangle$. This means that

$$\hat{a}^\dagger_{p_1}\hat{a}^\dagger_{p_2} = \lambda\hat{a}^\dagger_{p_2}\hat{a}^\dagger_{p_1}, \tag{3.11}$$

where λ is a constant. We assume two possibilities,[6] $\lambda = \pm 1$, and they correspond to wave functions which are either symmetric or antisymmetric under particle exchange. The particles are then called **bosons** or **fermions** and we will find that the two cases correspond to two different types of commutation relation for the creation and annihilation operators. We will examine each case in turn.

Case I: $\lambda = 1$, *bosons:* In this case

$$\hat{a}^\dagger_{p_2}\hat{a}^\dagger_{p_1} = \hat{a}^\dagger_{p_1}\hat{a}^\dagger_{p_2}. \tag{3.12}$$

[6] In Chapter 29 we examine the possibility of other choices.

"Anyons?" recently discovered in condensed matter !

Case II: $\lambda = -1$, *fermions:* In this case we'll write the fermion creation operators $\hat{c}^\dagger_i$ to distinguish them from the boson operators (written $\hat{a}^\dagger_i$). Plugging in $\lambda = -1$, we find that

$$\left\{\hat{c}^\dagger_i, \hat{c}^\dagger_j\right\} \equiv \hat{c}^\dagger_i\hat{c}^\dagger_j + \hat{c}^\dagger_j\hat{c}^\dagger_i = 0, \tag{3.19}$$

where the curly brackets indicate an **anticommutator**. Fermion operators anticommute: when you change their order you pick up a minus sign since $\hat{c}^\dagger_i\hat{c}^\dagger_j = -\hat{c}^\dagger_j\hat{c}^\dagger_i$. In particular, setting $i = j$ we find

$$\hat{c}^\dagger_i\hat{c}^\dagger_i + \hat{c}^\dagger_i\hat{c}^\dagger_i = 0 \quad \text{and hence} \quad \hat{c}^\dagger_i\hat{c}^\dagger_i = 0. \tag{3.20}$$

Peskin Why Fermions-Pauli

- **Peskin: START w/ regular commutation relations for contractions as in previous cases:** (infinities aside)
- ...Canonical Quantization...
- **Resulting H can indefinitely create more and more negative energies** Fixed by imposing anti-commutation relation

We are now ready to write H in terms of the a's and b's. After another short calculation (making use of the orthogonality relations (3.60), (3.63), and (3.65)), we find

$$H = \int \frac{d^3p}{(2\pi)^3} \sum_s \left(E_{\mathbf{p}} a^{s\dagger}_{\mathbf{p}} a^s_{\mathbf{p}} - E_{\mathbf{p}} b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} \right). \tag{3.90}$$

Something is terribly wrong with the second term: By creating more and more particles with $b^\dagger$, we can lower the energy indefinitely. (It would not have helped to rename $b \leftrightarrow b^\dagger$, since doing so would ruin the commutation relation (3.89).)

Why Anti-commute/Why Fermi stats!!!

- Feynman himself quoted on this:

> An elementary explanation for the spin-statistics theorem cannot be given despite the fact that the theorem is so simple to state. In the Feynman Lectures on Physics, Richard Feynman said that this probably means that we do not have a complete understanding of the fundamental principle involved. see **Further reading** below.

- Probably a better way to construct field theory?

- But computers are basically allowing us not to really be limited by the non-simpleness

More QED

- After Trace steps for simplifying gamma matrices

Another example: ee+-->mumu+

- Start w/ Griff: (but for mu – same for e^+e^- t-channel)
  - worked out already in griff
- Let's do s-channel diagram (annihilation)

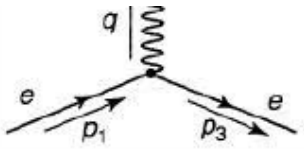


Fig. 7.4 Electron–positron scattering.

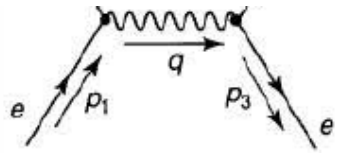


Fig. 7.5 Second diagram contributing to electron–positron scattering.

is thus

$$\mathcal{M}_1 = -\frac{g_e^2}{(p_1 - p_3)^2}[\bar{u}(3)\gamma^\mu u(1)][\bar{v}(2)\gamma_\mu v(4)] \qquad (7.108)$$

The other diagram represents virtual annihilation of the electron and positron, followed by pair production (Figure 7.5):

$$(2\pi)^4 \int [\bar{u}(3)(ig_e\gamma^\mu)v(4)]\frac{-ig_{\mu\nu}}{q^2}[\bar{v}(2)(ig_e\gamma^\nu)u(1)] \times \delta^4(q - p_3 - p_4)\delta^4(p_1 + p_2 - q)\,d^4q$$

$$\frac{g^2}{s}\ \bar{u}_3\gamma^\mu v_4\ \bar{v}_2\gamma_\mu v_1\ \bar{u}_3\gamma^\mu v_4\ \bar{v}_2\gamma_\mu v_1$$

Continuing annihil

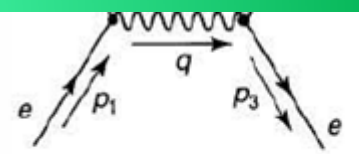


Fig. 7.5 Second diagram contributing to electron–positron scattering.

$$\frac{g^2}{s}\; \bar{u}_3 \gamma^\mu v_4\; \bar{v}_2 \gamma_\mu u_1 \left(\bar{u}_3 \gamma^\nu v_4\right)^* \left(\bar{v}_2 \gamma_\nu u_1\right)^*$$

$$\bar{u}_3 \gamma^\mu v_4\; \bar{v}_2 \gamma_\mu u_1\; \bar{v}_4 \gamma^\nu u_3\; \bar{u}_1 \gamma_\nu v_2$$

$$\bar{u}_3 \gamma^\mu v_4\; \bar{v}_4 \gamma^\nu u_3\; \bar{v}_2 \gamma_\mu u_1\; \bar{u}_1 \gamma_\nu v_2$$

$\not{p}_4 + \not{r}_2$ $\qquad$ $\not{p}_1$

Completeness

Continuing annihil

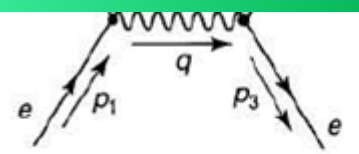


Fig. 7.5 Second diagram contributing to electron–positron scattering.

$$\frac{g^2}{s}\ \bar{u}_3\gamma^\mu v_4\ \bar{v}_2\gamma_\mu u_1\left(\bar{u}_3\gamma^\nu v_4\right)^*\left(\bar{v}_2\gamma_\nu u_1\right)^*$$

$$\bar{u}_3\gamma^\mu v_4\ \bar{v}_2\gamma_\mu u_1\ \bar{v}_4\gamma^\nu u_3\ \bar{u}_1\gamma_\nu v_2$$

$$\bar{u}_3\gamma^\mu v_4\ \bar{v}_4\gamma^\nu u_3\ \bar{v}_2\gamma_\mu u_1\ \bar{u}_1\gamma_\nu v_2$$

$\not{p}_4 + \not{r}_2$ $\qquad$ $\not{p}_1$

Completeness

Continuing annihil

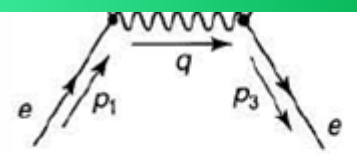


Fig. 7.5 Second diagram contributing to electron–positron scattering.

$$\frac{g^2}{s}\;\bar{u}_3\gamma^\mu v_4\;\bar{v}_2\gamma_\mu u_1\left(\bar{u}_3\gamma^\nu v_4\right)^*\left(\bar{v}_2\gamma_\nu u_1\right)^*$$

$$\bar{u}_3\gamma^\mu v_4\;\bar{v}_2\gamma_\mu u_1\;\bar{v}_4\gamma^\nu u_3\;\bar{u}_1\gamma_\nu v_2$$

Rearrange

$$\bar{u}_3\gamma^\mu v_4\;\bar{v}_4\gamma^\nu u_3\;\bar{v}_2\gamma_\mu u_1\;\bar{u}_1\gamma_\nu v_2$$

$$\not{p}_4 + m_e \qquad \not{p}_1 \qquad \leftarrow \text{Completeness}$$

$$\bar{u}_3\gamma^\mu\not{p}_4\gamma^\nu u_3\;\bar{v}_2\gamma_\mu\not{p}_1\gamma_\nu v_2$$

$$\mathrm{Tr}\left(\gamma^\mu\not{p}_4\gamma^\nu\not{p}_3\right)\;\mathrm{Tr}\left(\gamma_\mu\not{p}_1\gamma_\nu\not{p}_2\right)$$

Trace zoom in

Need to evaluate $\sum_{a-spins}\left(\sum_{b-spins} G\right) = \sum_{s_a=1,2} \overline{u}^{(s_a)}(p_a) Q u^{(s_a)}(p_a)$

Write out the matrix multiplication explicitly:

$$\sum_{i,j=1,4}\sum_{s_a=1,2} \overline{u}^{(s_a)}(p_a)_i Q_{ij} u^{(s_a)}(p_a)_j = \sum_{i,j=1,4} Q_{ij}\left\{\sum_{s_a=1,2} u^{(s_a)}(p_a)\overline{u}^{(s_a)}(p_a)\right\}_{ji}$$

[see footnote on pg 251 of text]

each of these 3 quantities is just a number

$$\sum_i\sum_j\sum_{s=1,2} \overline{u}_i^{(s)} Q_{ij} u_j^{(s)} \Rightarrow \sum_i\sum_j Q_{ij}\sum_{s=1,2} \overline{u}_i^{(s)} u_j^{(s)}$$

using $\overline{u}_i u_j = \left(u^\dagger \gamma^0\right)_i u_j = \left(u\overline{u}\right)_{ji}$ $\quad \sum_{i,j} Q_{ij}\left\{\sum_{s=1,2} \overline{u}_i^{(s)} u_j^{(s)}\right\} = \sum_{i,j} Q_{ij}\left\{\sum_{s=1,2} u^{(s)}\overline{u}^{(s)}\right\}_{ji}$

so $\sum_{a-spins}\left(\sum_{b-spins} G\right) = \sum_{i,j=1,4} Q_{ij}\left\{\sum_{s_a=1,2} u^{(s_a)}(p_a)\overline{u}^{(s_a)}(p_a)\right\}_{ji} = \sum_{i,j=1,4} Q_{ij}\left(p_a + m_a c\right)_{ji} = Tr\left(Q\left(p_a + m_a c\right)\right)$

To summarize Tr appearance

$$\sum_{i,j=1,4} \sum_{s_a=1,2} \overline{u}^{(s_a)}(p_a)_i Q_{ij} u^{(s_a)}(p_a)_j$$

$$= Tr\left(Q\left(p_a + m_a c\right)\right)$$

Continuing

- This form results in TWO traces multiplied w/ ~same form

I recommend changing indicies to match Griff identities for complicated ones

$$\bar{u}_3 \gamma^\mu \not{p}_4 \gamma^\nu u_3 \; \bar{v}_2 \gamma_\mu \not{p}_1 \gamma_\nu v_2$$

$$\mathrm{Tr}\left(\gamma^\mu \not{p}_4 \gamma^\nu \not{p}_3\right) \mathrm{Tr}\left(\gamma_\mu \not{p}_1 \gamma_\nu \not{p}_2\right)$$

13. $\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\lambda\gamma^\sigma)$

$= 4(g^{\mu\nu}g^{\lambda\sigma} - g^{\mu\lambda}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\lambda})$

13'. $\mathrm{Tr}(\not{a}\not{b}\not{c}\not{d})$

$= 4(a\cdot b\, c\cdot d - a\cdot c\, b\cdot d + a\cdot d\, b\cdot c)$

$\gamma^\mu \gamma^\nu p_{4\nu} \gamma^\lambda \gamma^\sigma p_{3\sigma}$

$\gamma^\mu \gamma^\nu \gamma^\lambda \gamma^\sigma \; p_{4\nu} p_{3\sigma}$

$$\mathrm{Tr}\left(p_4^\mu p_3^\lambda + p_4^\lambda p_3^\mu - p_4 \cdot p_3 g^{\mu\lambda}\right)$$

Cool. Now Use the form derived for left now for right hand Trace

Continuing

- This form results in TWO traces multiplied w/ ~same form

$$\bar{u}_3 \gamma^\mu \not{p}_4 \gamma^\nu u_3 \quad \bar{v}_2 \gamma_\mu \not{p}_1 \gamma_\nu v_2$$

$$\mathrm{Tr}(\gamma^\mu \not{p}_4 \gamma^\nu \not{p}_3)\ \mathrm{Tr}(\gamma_\mu \not{p}_1 \gamma_\nu \not{p}_2)$$

13. $\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\lambda\gamma^\sigma)$

$= 4(g^{\mu\nu}g^{\lambda\sigma} - g^{\mu\lambda}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\lambda})$

13'. $\mathrm{Tr}(\not{a}\not{b}\not{c}\not{d})$

$= 4(a\cdot b\, c\cdot d - a\cdot c\, b\cdot d + a\cdot d\, b\cdot c)$

$\gamma^\mu\gamma^\nu p_{4\nu}\gamma^\lambda\gamma^\sigma p_{3\sigma}$

$\gamma^\mu\gamma^\nu\gamma^\lambda\gamma^\sigma\, p_{4\nu}p_{3\sigma}$

$\left(p_4^\mu p_3^\lambda + p_4^\lambda p_3^\mu - p_4\cdot p_3\, g^{\mu\lambda}\right)$

$p_{1\mu}p_{2\lambda} + p_{1\lambda}p_{2\mu} - p_1\cdot p_2\, g_{\mu\lambda}$

Continuing

$$\bar{u}_3 \gamma^\mu \not{p}_4 \gamma^\nu u_3 \; \bar{v}_2 \gamma_\mu \not{p}_1 \gamma_\nu v_2$$

$$\mathrm{Tr}(\gamma^\mu \not{p}_4 \gamma^\nu \not{p}_3)\,\mathrm{Tr}(\gamma_\mu \not{p}_1 \gamma_\nu \not{p}_2)$$

13. $\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\lambda\gamma^\sigma)$

$= 4(g^{\mu\nu}g^{\lambda\sigma} - g^{\mu\lambda}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\lambda})$

13′. $\mathrm{Tr}(\not{a}\not{b}\not{c}\not{d})$

$= 4(a\cdot b\, c\cdot d - a\cdot c\, b\cdot d + a\cdot d\, b\cdot c)$

$\gamma^\mu\gamma^\nu\gamma^\lambda\gamma^\sigma\, p_{4\nu} p_{3\sigma}$

$4\left(p_4^\mu p_3^\lambda + p_3^\mu p_4^\lambda - g^{\mu\lambda} p_3\cdot p_4\right)$

S

Trace Theorems/Casimir's Trick

atrices
untless
ors

→

8. $\gamma_\mu \gamma^\nu \gamma^\lambda \gamma^\mu = 4g^{\nu\lambda}$ 8′. $\gamma_\mu \not{a} \not{b} \gamma^\mu = 4(a \cdot b)$

9. $\gamma_\mu \gamma^\nu \gamma^\lambda \gamma^\sigma \gamma^\mu = -2\gamma^\sigma \gamma^\lambda \gamma^\nu$ 9′. $\gamma_\mu \not{a} \not{b} \not{c} \gamma^\mu = -2\not{c}\not{b}\not{a}$

nd a collection of 'trace theorems':

10. The trace of the product of an *odd* number of gamma matrices is *zero*.

11. $\mathrm{Tr}(1) = 4$

12. $\mathrm{Tr}(\gamma^\mu \gamma^\nu) = 4g^{\mu\nu}$ 12′. $\mathrm{Tr}(\not{a}\not{b}) = 4(a \cdot b)$

13. $\mathrm{Tr}(\gamma^\mu \gamma^\nu \gamma^\lambda \gamma^\sigma) = 4(g^{\mu\nu}g^{\lambda\sigma} - g^{\mu\lambda}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\lambda})$ 13′. $\mathrm{Tr}(\not{a}\not{b}\not{c}\not{d}) = 4(a \cdot b\, c \cdot d - a \cdot c\, b \cdot d + a \cdot d\, b \cdot c)$

1. $\mathrm{Tr}(A + B) = \mathrm{Tr}(A) + \mathrm{Tr}(B)$
2. $\mathrm{Tr}(\alpha A) = \alpha\, \mathrm{Tr}(A)$
3. $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$

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4. $g_{\mu\nu}g^{\mu\nu} = 4$
5. $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}$ 5′. $\not{a}\not{b} + \not{b}\not{a} = 2a \cdot b$
6. $\gamma_\mu\gamma^\mu = 4$
7. $\gamma_\mu\gamma^\nu\gamma^\mu = -2\gamma^\nu$ 7′. $\gamma_\mu \not{a} \gamma^\mu = -2\not{a}$

---

roduct of an *even* number of γ matrices, it
r $(\gamma^5\gamma^\mu\gamma^\nu\gamma^\lambda) = 0$. When γ^5 is multiplied

15′. $\mathrm{Tr}(\gamma^5 \not{a}\not{b}) = 0$

$\lambda\sigma$ 16′. $\mathrm{Tr}(\gamma^5 \not{a}\not{b}\not{c}\not{d}) = 4i\epsilon^{\mu\nu\lambda\sigma} a_\mu b_\nu c_\lambda d_\sigma$

Continuing annih

- Two

$$4\left(p_4^\mu p_3^\lambda + p_3^\mu p_4^\lambda - g^{\mu\lambda} p_3 \cdot p_4\right)$$

S

Continuing annih

- (Change indices back)

$$4\left(p_4^\mu p_3^\lambda + p_3^\mu p_4^\lambda - g^{\mu\lambda} p_3\cdot p_4\right) \qquad \text{all} \quad p_1, p_2$$

$$S \qquad p_1\cdot p_2 = s$$

$$4^2\left(p_4^\mu p_3^\nu + p_3^\nu p_4^\mu - g^{\mu\nu} s\right)\left(p_{1\mu} p_{2\nu} + p_{1\nu} p_{2\mu} - g_{\mu\nu} s\right)$$

Continuing annih

- Two

$$4\left(p_4^\mu p_3^\lambda + p_3^\mu p_4^\lambda - g^{\mu\lambda} p_3\cdot p_4\right) \qquad \text{all } p_1, p_2$$

$$S$$

from propagator

$$4^2\left(p_4^\mu p_3^\nu + p_3^\mu p_4^\nu - g^{\mu\nu} S\right)\left(p_{1\mu} p_{2\nu} + p_{1\nu} p_{2\mu} - g_{\mu\nu} S\right) \qquad p_1\cdot p_5 = S$$

$$\frac{16^4}{S^2}\Big[6\left(p_4\cdot p_1\,(p_3 p_2) + (p_3\cdot p_1)\, p_4\cdot p_2 - p_4\cdot p_3\, S\right)$$

$u \quad u \qquad t \quad t \qquad (s(s)$

c

+ 6 other terms

Continuing annih

- Two

$$4\left(p_4^\mu p_3^\lambda + p_3^\mu p_4^\lambda - g^{\mu\lambda} p_3\cdot p_4\right) \qquad \text{all } p_1, p_2$$

$$S$$

$p_1 \cdot p_2 = s$

from propagator

$$4^2\left(p_4^\mu p_3^\nu + p_3^\mu p_4^\nu - g^{\mu\nu} s\right)\left(p_{1\mu} p_{2\nu} + p_{1\nu} p_{2\mu} - g_{\mu\nu} s\right)$$

$$\frac{1}{s^2}\, 16\left(p_4\cdot p_1 (p_3 p_2) + (p_1\cdot p_3)(p_4\cdot p_2) - p_4\cdot p_3\, s\right)$$

$u \quad u \qquad t \quad t \qquad (s(s)$

$$(p_4\cdot p_2)(p_3\cdot p_1) + (p_3\cdot p_2)(p_1\cdot p_4) - \cdot s^2$$

$$t \quad + \quad t \quad + s^2 \quad + s^2 \quad + 4s^2$$

ee+->mumu+ Help

- Start w/ Griff: (but for mu)
- what I meant the whole problem is ONLY ths diagram

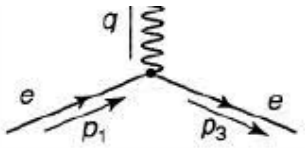


Fig. 7.4 Electron–positron scattering.

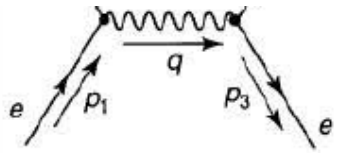


Fig. 7.5 Second diagram contributing to electron–positron scattering.

is thus

$$\mathscr{M}_1 = -\frac{g_e^2}{(p_1 - p_3)^2}[\bar{u}(3)\gamma^\mu u(1)][\bar{v}(2)\gamma_\mu v(4)] \tag{7.108}$$

The other diagram represents virtual annihilation of the electron and positron, followed by pair production (Figure 7.5):

$$(2\pi)^4 \int [\bar{u}(3)(ig_e\gamma^\mu)v(4)]\frac{-ig_{\mu\nu}}{q^2}[\bar{v}(2)(ig_e\gamma^\nu)u(1)]$$
$$\times \delta^4(q - p_3 - p_4)\delta^4(p_1 + p_2 - q)\,d^4q$$

$$\frac{g^2}{s}\ \bar{u}_3\gamma^\mu v_4\ \bar{v}_2\gamma_\mu v_1 \left(\bar{u}_3\gamma^\mu v_4\right)^* \left(\bar{v}_2\gamma_\mu v_1\right)^*$$

Continuing annihil

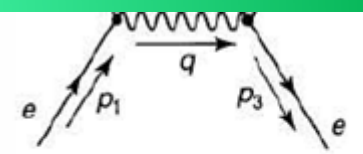


Fig. 7.5 Second diagram contributing to electron–positron scattering.

$$\frac{\mathcal{B}^2}{S}\quad \bar{u}_3\gamma^\mu v_4\ \bar{v}_2\gamma_\mu u_1\left(\bar{u}_3\gamma^\nu v_4\right)^*\left(\bar{v}_2\gamma_\nu u_1\right)^*$$

$$\bar{u}_3\gamma^\mu v_4\ \bar{v}_2\gamma_\mu u_1\ \bar{v}_4\gamma^\nu u_3\ \bar{u}_1\gamma_\nu v_2$$

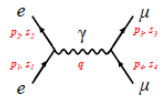
$$\bar{u}_3\gamma^\mu v_4\ \bar{v}_4\gamma^\nu u_3\ \bar{v}_2\gamma_\mu u_1\ \bar{u}_1\gamma_\nu v_2$$

$$\bar{u}_3\gamma^\mu \not{p}_4\gamma^\nu u_3\ \bar{v}_2\gamma_\mu \not{p}_1\gamma_\nu v_2$$

A couple looks at the e+e-→mu+mu-

- In class we looked through the long long calcs reproducing the Griff results for e+e-→mu+mu (from Krieger .. much more complicated than Griff short results would seem to indicate.) but see next slide

Amplitude $\mathcal{M}$ for the process $e^+e^-\to\mu^+\mu^-$



Here we need antiparticle spinors for the first time.

$$\int \underbrace{\left[\bar{v}(2)\,ig\gamma^{\mu}\,u(1)(2\pi)^4\delta^4(p_1+p_2-q)\right]}_{\text{electron line}}\frac{-ig_{\mu\nu}}{q^2}\underbrace{\left[\bar{u}(3)\,ig\gamma^{\nu}\,v(4)(2\pi)^4\delta^4(q-p_3-p_4)\right]}_{\text{muon line}}\frac{d^4q}{(2\pi)^4}$$

$$=-\frac{g^2}{(p_1+p_2)^2}\left[\bar{v}(2)\gamma^{\mu}u(1)\right]\left[\bar{u}(3)\gamma_{\mu}v(4)\right]\quad |\mathcal{M}|^2=\left[\frac{g^2}{(p_1+p_2)^2}\right]^2\left[\bar{v}(2)\gamma^{\mu}u(1)\right]\left[\bar{u}(3)\gamma_{\mu}v(4)\right]^*\left[\bar{u}(3)\gamma_{\nu}v(4)\right]^*$$

and that the sign of the mass term changes in the case of anti-particle spinors.

Should be anti-particle spinor

$$\sum_{\text{all spins}}\left[\bar{u}(3)\gamma_{\mu}v(4)\right]\left[\bar{u}(3)\gamma_{\nu}u(4)\right]^*=Tr\left(\gamma_{\mu}\left(p_4-Mc\right)\gamma^0\gamma_{\nu}^{\dagger}\gamma^0\left(p_3+Mc\right)\right)=Tr\left(\gamma_{\mu}\left(p_4-Mc\right)\gamma_{\nu}\left(p_3+Mc\right)\right)$$

minus sign b.c. antiparticle v completeness

Remember, we average over incoming spins and sum over outgoing spins:

$$\left\langle|\mathcal{M}|^2\right\rangle=\frac{1}{4}\frac{g^4}{(p_1+p_2)^4}\underbrace{Tr\left(\gamma^{\mu}(p_1+mc)\gamma^{\nu}(p_2-mc)\right)Tr\left(\gamma_{\mu}(p_4-Mc)\gamma_{\nu}(p_3+Mc)\right)}$$

Look at trace part:

$$=\left[Tr\left(\gamma^{\mu}p_1\gamma^{\nu}p_2\right)-(mc)^2Tr\left(\gamma^{\mu}\gamma^{\nu}\right)\right]\left[Tr\left(\gamma_{\mu}p_4\gamma_{\nu}p_3\right)-(Mc)^2Tr\left(\gamma_{\mu}\gamma_{\nu}\right)\right]$$

since terms with three γ matrices have Trace = 0 [Rule 10] and $Tr(\gamma^{\mu}\gamma^{\nu})=4g^{\mu\nu}$

$$\not{p}_1=p_{1\lambda}\gamma^{\lambda}\qquad \not{p}_2=p_{2\sigma}\gamma^{\sigma}\qquad\qquad \not{p}_4=p_{4\lambda}\gamma^{\lambda}\qquad \not{p}_3=p_{3\sigma}\gamma^{\sigma}$$

$$=\left[p_{1\lambda}p_{2\sigma}\underbrace{Tr\left(\gamma^{\mu}\gamma^{\lambda}\gamma^{\nu}\gamma^{\sigma}\right)}_{\text{rule 13}}-4(mc)^2g^{\mu\nu}\right]\left[p^{4\lambda}p^{3\sigma}\underbrace{Tr\left(\gamma_{\mu}\gamma_{\lambda}\gamma_{\nu}\gamma_{\sigma}\right)}_{\text{rule 13}}-4(Mc)^2g_{\mu\nu}\right]$$

$$=\left[4p_{1\lambda}p_{2\sigma}\left(g^{\mu\lambda}g^{\nu\sigma}-g^{\mu\nu}g^{\lambda\sigma}+g^{\mu\sigma}g^{\lambda\nu}\right)-4(mc)^2g^{\mu\nu}\right]\left[4p^{4\lambda}p^{3\sigma}\left(g_{\mu\lambda}g_{\nu\sigma}-g_{\mu\nu}g_{\lambda\sigma}+g_{\mu\sigma}g_{\lambda\nu}\right)-4(Mc)^2g_{\mu\nu}\right]$$

$$=16\left[p_1^{\mu}p_2^{\nu}-(p_1\cdot p_2)g^{\mu\nu}+p_2^{\mu}p_1^{\nu}-(mc)^2g^{\mu\nu}\right]\left[p_{4\mu}p_{3\nu}-(p_4\cdot p_3)g_{\mu\nu}+p_{3\mu}p_{4\nu}-(Mc)^2g_{\mu\nu}\right]$$

$$=16\Big[\;(p_1\cdot p_4)(p_2\cdot p_3)-(p_1\cdot p_2)(p_4\cdot p_3)+(p_1\cdot p_3)(p_2\cdot p_4)-(p_1\cdot p_2)(Mc)^2$$
$$-(p_1\cdot p_2)(p_3\cdot p_4)+4(p_1\cdot p_2)(p_4\cdot p_3)-(p_1\cdot p_2)(p_3\cdot p_4)+4(p_1\cdot p_2)(Mc)^2$$
$$+(p_2\cdot p_4)(p_1\cdot p_3)-(p_1\cdot p_2)(p_4\cdot p_3)+(p_2\cdot p_3)(p_1\cdot p_4)-(p_1\cdot p_2)(Mc)^2$$
$$-(mc)^2(p_4\cdot p_3)+4(mc)^2(p_4\cdot p_3)-(mc)^2(p_3\cdot p_4)+4(mc)^2(Mc)^2\;\Big]$$

$$=16\left[2(p_1\cdot p_3)(p_2\cdot p_4)+2(p_1\cdot p_4)(p_2\cdot p_3)+2(Mc)^2(p_1\cdot p_2)+2(mc)^2(p_3\cdot p_4)+4(mc)^2(Mc)^2\right]$$

$$=32\left[(p_1\cdot p_3)(p_2\cdot p_4)+(p_1\cdot p_4)(p_2\cdot p_3)+(Mc)^2(p_1\cdot p_2)+(mc)^2(p_3\cdot p_4)+2(mc)^2(Mc)^2\right]$$

$$\left\langle|\mathcal{M}|^2\right\rangle=\frac{32}{4}\frac{g^4}{(p_1+p_2)^4}\left[(p_1\cdot p_3)(p_2\cdot p_4)+(p_1\cdot p_4)(p_2\cdot p_3)+(Mc)^2(p_1\cdot p_2)+(mc)^2(p_3\cdot p_4)+2(mc)^2(Mc)^2\right]$$

$$\left\langle|\mathcal{M}|^2\right\rangle=8\left(\frac{g^2}{(p_1+p_2)^2}\right)^2\left[(p_1\cdot p_3)(p_2\cdot p_4)+(p_1\cdot p_4)(p_2\cdot p_3)+(Mc)^2(p_1\cdot p_2)+(mc)^2(p_3\cdot p_4)+2(mc)^2(Mc)^2\right]$$

Compare eqn. 7.129 for $e\mu\to e\mu$ with $q^2=(p_1+p_2)^2$ and p_2 swapped with p_3.

$$\left\langle|\mathcal{M}|^2\right\rangle=g^4\left\{1+\cos^2\theta-\frac{(mc)^2\cos^2\theta}{(E/c)^2}-\frac{(Mc)^2\cos^2\theta}{(E/c)^2}+\frac{(mc)^2(Mc)^2}{(E/c)^4}+\frac{(mc)^2}{(E/c)^2}+\frac{(Mc)^2}{(E/c)^2}\right\}$$

$$=g^4\left\{1+\frac{(mc^2)^2}{E^2}+\frac{(Mc^2)^2}{E^2}+\left[1-\frac{(mc^2)^2}{E^2}-\frac{(Mc^2)^2}{E^2}+\frac{(mc^2)^2(Mc^2)^2}{E^4}\right]\cos^2\theta\right\}$$

$$=g^4\left\{1+\frac{(mc^2)^2}{E^2}+\frac{(Mc^2)^2}{E^2}+\left[1-\left(\frac{mc^2}{E}\right)^2\right]\left[1-\left(\frac{Mc^2}{E}\right)^2\right]\cos^2\theta\right\}$$ [Griffiths 8.4]

Using $\frac{d\sigma}{d\Omega}=\left(\frac{\hbar c}{8\pi}\right)^2\frac{\left\langle|\mathcal{M}|^2\right\rangle}{(2E)^2}\frac{|\vec{p}_f|}{|\vec{p}_i|}$ [Griffiths 6.42] and $\frac{|\vec{p}_f|}{|\vec{p}_i|}=\frac{\sqrt{1-(Mc^2/E)^2}}{\sqrt{1-(mc^2/E)^2}}$ [basic kinematics] try it

yields the differential cross-section. For the total cross-section, need to integrate over θ, φ.

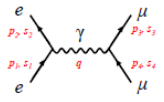
Integrate $\left\langle|\mathcal{M}|^2\right\rangle$ (from above) since that contains all the angular dependence:

$$=2\pi g^4\int\left\{1+\frac{(mc^2)^2}{E^2}+\frac{(Mc^2)^2}{E^2}+\left[1-\left(\frac{mc^2}{E}\right)^2\right]\left[1-\left(\frac{Mc^2}{E}\right)^2\right]\cos^2\theta\right\}\sin\theta d\theta$$

A couple looks at the e+e-→mu+mu-

- Here are calcs e+e-→mu+mu (from Krieger .. much more complicated than Griff short results would seem to indicate.) but see next slide

Amplitude $\mathcal{M}$ for the process $e^+e^-\rightarrow\mu^+\mu^-$



Here we need antiparticle spinors for the first time.

$$\int \underbrace{\left[\bar{v}(2)\, ig\gamma^{\mu}\, u(1)(2\pi)^4\delta^4(p_1+p_2-q)\right]}_{\text{electron line}}\frac{-ig_{\mu\nu}}{q^2}\underbrace{\left[\bar{u}(3)\, ig\gamma^{\nu}\, v(4)(2\pi)^4\delta^4(q-p_3-p_4)\right]}_{\text{muon line}}\frac{d^4q}{(2\pi)^4}$$

$$= -\frac{g^2}{(p_1+p_2)^2}\left[\bar{v}(2)\gamma^{\mu}u(1)\right]\left[\bar{u}(3)\gamma_{\mu}v(4)\right] \quad |\mathcal{M}|^2 = \left[\frac{g^2}{(p_1+p_2)^2}\right]^2\left[\bar{v}(2)\gamma^{\mu}u(1)\right]\left[\bar{u}(3)\gamma_{\mu}v(4)\right]\left[\bar{v}(2)\gamma^{\nu}u(1)\right]^*\left[\bar{u}(3)\gamma_{\nu}v(4)\right]^*$$

minus sign b.c. antiparticle v completeness

and that the sign of the mass term changes in the case of anti-particle spinors.

Should be anti-particle spinor

$$\sum_{\text{all spins}}\left[\bar{u}(3)\gamma_{\mu}v(4)\right]\left[\bar{u}(3)\gamma_{\nu}u(4)\right]^* = Tr\left(\gamma_{\mu}\left(p_4-Mc\right)\gamma^0\gamma_{\nu}^{\dagger}\gamma^0\left(p_3+Mc\right)\right) = Tr\left(\gamma_{\mu}\left(p_4-Mc\right)\gamma_{\nu}\left(p_3+Mc\right)\right)$$

A couple looks at the e+e-→mu+mu-

Remember, we average over incoming spins and sum over outgoing spins:

$$\left\langle \left|\mathcal{M}\right|^2 \right\rangle = \frac{1}{4}\frac{g^4}{\left(p_1+p_2\right)^4} \underbrace{Tr\left(\gamma^\mu\left(p_1+mc\right)\gamma^\nu\left(p_2-mc\right)\right)Tr\left(\gamma_\mu\left(p_4-Mc\right)\gamma_\nu\left(p_3+Mc\right)\right)}$$

Look at trace part:

$$= \left[Tr\left(\gamma^\mu p_1 \gamma^\nu p_2\right) - \left(mc\right)^2 Tr\left(\gamma^\mu\gamma^\nu\right)\right]\left[Tr\left(\gamma_\mu p_4 \gamma_\nu p_3\right) - \left(Mc\right)^2 Tr\left(\gamma_\mu\gamma_\nu\right)\right]$$

since terms with three γ matrices have Trace = 0 [Rule 10] and $Tr(\gamma^\mu\gamma^\nu)=4g^{\mu\nu}$

$$\not{p}_1 = p_{1\lambda}\gamma^\lambda \quad \not{p}_2 = p_{1\sigma}\gamma^\sigma \qquad\qquad \not{p}_4 = p_{4\lambda}\gamma^\lambda \quad \not{p}_3 = p_{3\sigma}\gamma^\sigma$$

$$= \left[p_{1\lambda}p_{2\sigma}\underbrace{Tr\left(\gamma^\mu\gamma^\lambda\gamma^\nu\gamma^\sigma\right)}_{\text{rule 13}} - 4\left(mc\right)^2 g^{\mu\nu}\right]\left[p^{4\lambda}p^{3\sigma}\underbrace{Tr\left(\gamma_\mu\gamma_\lambda\gamma_\nu\gamma_\sigma\right)}_{\text{rule 13}} - 4\left(Mc\right)^2 g_{\mu\nu}\right]$$

$$= \left[4p_{1\lambda}p_{2\sigma}\left(g^{\mu\lambda}g^{\nu\sigma} - g^{\mu\nu}g^{\lambda\sigma} + g^{\mu\sigma}g^{\lambda\nu}\right) - 4\left(mc\right)^2 g^{\mu\nu}\right]\left[4p^{4\lambda}p^{3\sigma}\left(g_{\mu\lambda}g_{\nu\sigma} - g_{\mu\nu}g_{\lambda\sigma} + g_{\mu\sigma}g_{\lambda\nu}\right) - 4\left(Mc\right)^2 g_{\mu\nu}\right]$$

$$= 16\left[p_1^\mu p_2^\nu - \left(p_1\cdot p_2\right)g^{\mu\nu} + p_2^\mu p_1^\nu - \left(mc\right)^2 g^{\mu\nu}\right]\left[p_{4\mu}p_{3\nu} - \left(p_4\cdot p_3\right)g_{\mu\nu} + p_{3\mu}p_{4\nu} - \left(Mc\right)^2 g_{\mu\nu}\right]$$

A couple looks at the e+e-→mu+mu-

$$16\left[p_1^\mu p_2^\nu-(p_1\cdot p_2)g^{\mu\nu}+p_2^\mu p_1^\nu-(mc)^2 g^{\mu\nu}\right]\left[p_{4\mu}p_{3\nu}-(p_4\cdot p_3)g_{\mu\nu}+p_{3\mu}p_{4\nu}-(Mc)^2 g_{\mu\nu}\right]$$

$$\begin{aligned}=16\Big[&(p_1\cdot p_4)(p_2\cdot p_3)-(p_1\cdot p_2)(p_4\cdot p_3)+(p_1\cdot p_3)(p_2\cdot p_4)-(p_1\cdot p_2)(Mc)^2\\&-(p_1\cdot p_2)(p_3\cdot p_4)+4(p_1\cdot p_2)(p_4\cdot p_3)-(p_1\cdot p_2)(p_3\cdot p_4)+4(p_1\cdot p_2)(Mc)^2\\&+(p_2\cdot p_4)(p_1\cdot p_3)-(p_1\cdot p_2)(p_4\cdot p_3)+(p_2\cdot p_3)(p_1\cdot p_4)-(p_1\cdot p_2)(Mc)^2\\&-(mc)^2(p_4\cdot p_3)+4(mc)^2(p_4\cdot p_3)-(mc)^2(p_3\cdot p_4)+4(mc)^2(Mc)^2\Big]\end{aligned}$$

$$=16\left[2(p_1\cdot p_3)(p_2\cdot p_4)+2(p_1\cdot p_4)(p_2\cdot p_3)+2(Mc)^2(p_1\cdot p_2)+2(mc)^2(p_3\cdot p_4)+4(mc)^2(Mc)^2\right]$$

$$=32\left[(p_1\cdot p_3)(p_2\cdot p_4)+(p_1\cdot p_4)(p_2\cdot p_3)+(Mc)^2(p_1\cdot p_2)+(mc)^2(p_3\cdot p_4)+2(mc)^2(Mc)^2\right]$$

$$\left\langle\left|\mathcal{M}\right|^2\right\rangle=\frac{32}{4}\frac{g^4}{(p_1+p_2)^4}\left[(p_1\cdot p_3)(p_2\cdot p_4)+(p_1\cdot p_4)(p_2\cdot p_3)+(Mc)^2(p_1\cdot p_2)+(mc)^2(p_3\cdot p_4)+2(mc)^2(Mc)^2\right]$$

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Compare eqn. 7.129 for $e\mu \rightarrow e\mu$ with $q^2=(p_1+p_2)^2$ and p_2 swapped with p_3.

e+e−

$$\left\langle |\mathcal{M}|^2 \right\rangle = g^4 \left\{ 1 + \cos^2\theta - \frac{(mc)^2 \cos^2\theta}{(E/c)^2} - \frac{(Mc)^2 \cos^2\theta}{(E/c)^2} + \frac{(mc)^2 (Mc)^2}{(E/c)^4} + \frac{(mc)^2}{(E/c)^2} + \frac{(Mc)^2}{(E/c)^2} \right\}$$

$$= g^4 \left\{ 1 + \frac{(mc^2)^2}{E^2} + \frac{(Mc^2)^2}{E^2} + \left[1 - \frac{(mc^2)^2}{E^2} - \frac{(Mc^2)^2}{E^2} + \frac{(mc^2)^2 (Mc^2)^2}{E^4} \right] \cos^2\theta \right\}$$

$$= g^4 \left\{ 1 + \frac{(mc^2)^2}{E^2} + \frac{(Mc^2)^2}{E^2} + \left[1 - \left(\frac{mc^2}{E} \right)^2 \right] \left[1 - \left(\frac{Mc^2}{E} \right)^2 \right] \cos^2\theta \right\} \quad \text{[Griffiths 8.4]}$$

Using $\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi} \right)^2 \frac{\left\langle |\mathcal{M}|^2 \right\rangle}{(2E)^2} \frac{|\vec{p}_f|}{|\vec{p}_i|}$ [Griffiths 6.42] and $\frac{|\vec{p}_f|}{|\vec{p}_i|} = \frac{\sqrt{1 - (Mc^2/E)^2}}{\sqrt{1 - (mc^2/E)^2}}$ [basic kinematics] try it

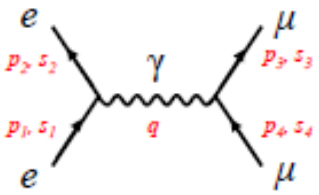
yields the differential cross-section. For the total cross-section, need to integrate over θ, φ.

Integrate $\left\langle |\mathcal{M}|^2 \right\rangle$ (from above) since that contains all the angular dependence:

$$= 2\pi g^4 \int \left\{ 1 + \frac{(mc^2)^2}{E^2} + \frac{(Mc^2)^2}{E^2} + \left[1 - \left(\frac{mc^2}{E} \right)^2 \right] \left[1 - \left(\frac{Mc^2}{E} \right)^2 \right] \cos^2\theta \right\} \sin\theta d\theta$$

Exercise

Amplitude $\mathcal{M}$ for the process $e^+e^- \rightarrow \mu^+\mu^-$



Here we need antiparticle spinors for the first time.

$$\int \underbrace{\left[\bar{v}(2)\, ig\gamma^{\mu}\, u(1)(2\pi)^4\delta^4(p_1+p_2-q)\right]}_{\text{electron line}}\frac{-ig_{\mu\nu}}{q^2}\underbrace{\left[\bar{u}(3)\, ig\gamma^{\nu}\, v(4)(2\pi)^4\delta^4(q-p_3-p_4)\right]}_{\text{muon line}}\frac{d^4q}{(2\pi)^4}$$

$$= -\frac{g^2}{(p_1+p_2)^2}\left[\bar{v}(2)\gamma^{\mu}u(1)\right]\left[\bar{u}(3)\gamma_{\mu}v(4)\right] \quad |\mathcal{M}|^2 = \left[\frac{g^2}{(p_1+p_2)^2}\right]^2\left[\bar{v}(2)\gamma^{\mu}u(1)\right]\left[\bar{u}(3)\gamma_{\mu}v(4)\right]\left[\bar{v}(2)\gamma^{\nu}u(1)\right]^*\left[\bar{u}(3)\gamma_{\nu}v(4)\right]^*$$

$$\left\langle|\mathcal{M}|^2\right\rangle = 8\left(\frac{g^2}{(p_1+p_2)^2}\right)^2\left[(p_1\cdot p_3)(p_2\cdot p_4)+(p_1\cdot p_4)(p_2\cdot p_3)+(Mc)^2(p_1\cdot p_2)+(mc)^2(p_3\cdot p_4)+2(mc)^2(Mc)^2\right]$$

- Write this in terms of Mendalstaam vars

$$\left\langle\right\rangle = 8\left(\frac{g^2}{(p_1+p_2)^2}\right)^2\left[(p_1\cdot p_3)(p_2\cdot p_4)+(p_1\cdot p_4)(p_2\cdot p_3)+(Mc)^2(p_1\cdot p_2)+(mc)^2(p_3\cdot p_4)+2(mc)^2(\right.$$

Last Griff vs Spinless QED

- Here's Griff, lets use Mandelstam vars to compare, identify Mandeltams...

$$\langle |\mathcal{M}|^2 \rangle = \frac{4g_e^4}{(p_1 - p_3)^4} \left\{ 2(p_1 \cdot p_2)(p_3 \cdot p_4) + 2(p_2 \cdot p_3)(p_1 \cdot p_4) - 2(p_1 \cdot p_3)(Mc)^2 - 2(p_2 \cdot p_4)(mc)^2 + 4(mMc^2)^2 \right\}$$

s s u u t t

(t^2)

so $M^2 \propto \dfrac{s^2 + u^2}{t^2}$ + t m² terms (negligible)

- Note this is just one term of M, if ee→ee there would be a crossed u term → this one indeed reduces to Rutherford also !!!! (just like we saw spinless QED did
- Again compare to H&M spinless QED result (start from red box)

$M \propto (s-u)/t \rightarrow M^2 \propto \dfrac{s^2 + u^2 - 2su}{t^2}$

$p_1 \cdot p_2 + p_1 \cdot p_4 + p_3 \cdot p + p_3 \cdot p_4$

$s - u$

$$T_{fi} = -i \int j_\mu^{(1)}(x) \left(-\frac{1}{q^2} \right) j_{(2)}^\mu(x)\, d^4x.$$

M !!!! (no Feyn. rules) Griff ABC only had middle factor $1/q^2$ ...now p's too

$$\mathfrak{M} = (ie(p_A + p_C)^\mu) \left(-i \frac{g_{\mu\nu}}{q^2} \right) (ie(p_B + p_D)^\nu).$$

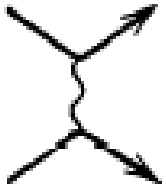
ALMOST EXACTLY SAME!!!!! they are, when relativistic – small diff is ~benefit of real QED also including spin magnetic dipole effect!!!!

Jibberish $\bar{u}\gamma u$'s are essentially momentum terms physically!!!

MUCH EASIER WAY: Halzen and Martin: Easy Switch of Crossing

- Use the symmetry of the other type of "crossing"
- --to first order it's equivalent to just switching the locations of the Mandelstam variables!!!!

$e^- \mu^- \rightarrow e^- \mu^-$



$\frac{s^2 + u^2}{t^2}$

(reminder next slide we did this in class

(Crossing $\downarrow s \leftrightarrow t$)

$e^- e^+ \rightarrow \mu^- \mu^+$



$\frac{u^2 + t^2}{s^2}$

- Note we've gotten rid of the 1/sin divergence (s is not angular dependent)
- We can integrate to get the full cross section

Exercise

- Prove relation 7:

1. $\mathrm{Tr}(A+B) = \mathrm{Tr}(A) + \mathrm{Tr}(B)$
2. $\mathrm{Tr}(\alpha A) = \alpha\,\mathrm{Tr}(A)$
3. $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$

---

4. $g_{\mu\nu}g^{\mu\nu} = 4$
5. $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}$ 5′. $\not{a}\not{b} + \not{b}\not{a} = 2a\cdot b$
6. $\gamma_\mu\gamma^\mu = 4$
7. $\gamma_\mu\gamma^\nu\gamma^\mu = -2\gamma^\nu$ 7′. $\gamma_\mu\not{a}\gamma^\mu = -2\not{a}$

Photon machinery

- Photon WF Solutions (solutions to Maxell d4Adx ∝ j

Plane-wave solutions with four-momentum $p = \left(\frac{E}{c}, \vec{p} \right)$ are of the form:

$$A^{\mu}(x) = a e^{-\frac{i}{\hbar} p \cdot x} \varepsilon^{\mu}(p)$$

Normalization factor (don't care here)

Polarization vector characterizing the photon spin-state.

$$\partial_{\mu}\partial^{\mu}A^{\nu} = 0 \quad \rightarrow \quad p_{\mu}p^{\mu} = 0 \quad \text{or} \quad E = |\vec{p}|c \quad \text{(correct for massless particle).}$$

The polarization vector ε^{μ} nominally has four components, but the Lorentz condition:

Feynman rules: incoming photon: $\varepsilon^{\mu}(p)$

outgoing photon: $\varepsilon^{\mu^*}(p)$

$$\left\{ \varepsilon^{\mu}_{(s)} \text{ for } s = 1,2 \right\}$$

Photon FIELDS QFT

- QFT : Spin 1 particles not spin 0
- But also Vectors this time real 4-v
- Somewhere in between spinors/dirac and Klein Gordon
- BUT MIXING OF 4D Components and Momentum!!!! MUCH SIMPLER THAN DIRAC

L/B:

The Lagrangian for this theory (**Step I**) is given by

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m^2 A_\mu A^\mu, \tag{13.17}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. This is just the Lagrangian for the electromagnetic field but with a Klein–Gordon-esque mass term on the right-hand side. Note that the sign of the second term that contains the factor m^2 is positive. In the Klein–Gordon equation the m^2 term was negative, but the reason for our sign choice here will become apparent in the following example.

Example 13.4

We can find the equations of motion for the field in this theory by plugging the Lagrangian (eqn 13.17) into the Euler–Lagrange equation. The result is that[9]

$$\partial_\mu F^{\mu\nu} + m^2 A^\nu = 0. \tag{13.18}$$

[9] Equation 13.18 can of course be written out purely in terms of A^μ as

$$\partial_\mu \left(\partial^\mu A^\nu - \partial^\nu A^\mu\right) + m^2 A^\nu = 0.$$

Photon FIELDS QFT

- Field operator expansion

$$\hat{A}^{\mu}(x) = \int \frac{\mathrm{d}^3 p}{(2\pi)^{\frac{3}{2}}} \frac{1}{(2E_{\boldsymbol{p}})^{\frac{1}{2}}} \left[\begin{pmatrix} \epsilon_1^0(p) \\ \epsilon_1^1(p) \\ \epsilon_1^2(p) \\ \epsilon_1^3(p) \end{pmatrix} \hat{a}_{\boldsymbol{p}1} \mathrm{e}^{-\mathrm{i}p\cdot x} + \begin{pmatrix} \epsilon_1^{0*}(p) \\ \epsilon_1^{1*}(p) \\ \epsilon_1^{2*}(p) \\ \epsilon_1^{3*}(p) \end{pmatrix} \hat{a}_{\boldsymbol{p}1}^{\dagger} \mathrm{e}^{\mathrm{i}p\cdot x} \right.$$
$$+ \begin{pmatrix} \epsilon_2^0(p) \\ \epsilon_2^1(p) \\ \epsilon_2^2(p) \\ \epsilon_2^3(p) \end{pmatrix} \hat{a}_{\boldsymbol{p}2} \mathrm{e}^{-\mathrm{i}p\cdot x} + \begin{pmatrix} \epsilon_2^{0*}(p) \\ \epsilon_2^{1*}(p) \\ \epsilon_2^{2*}(p) \\ \epsilon_2^{3*}(p) \end{pmatrix} \hat{a}_{\boldsymbol{p}2}^{\dagger} \mathrm{e}^{\mathrm{i}p\cdot x}$$
$$\left. + \begin{pmatrix} \epsilon_3^0(p) \\ \epsilon_3^1(p) \\ \epsilon_3^2(p) \\ \epsilon_3^3(p) \end{pmatrix} \hat{a}_{\boldsymbol{p}3} \mathrm{e}^{-\mathrm{i}p\cdot x} + \begin{pmatrix} \epsilon_3^{0*}(p) \\ \epsilon_3^{1*}(p) \\ \epsilon_3^{2*}(p) \\ \epsilon_3^{3*}(p) \end{pmatrix} \hat{a}_{\boldsymbol{p}3}^{\dagger} \mathrm{e}^{\mathrm{i}p\cdot x} \right], \qquad (13.24)$$

[16] The commutation relations between the creation and annihilation operators are

$$\left[\hat{a}_{\boldsymbol{p}\lambda}, \hat{a}_{\boldsymbol{q}\lambda'}^{\dagger}\right] = \delta^{(3)}(\boldsymbol{p}-\boldsymbol{q})\delta_{\lambda\lambda'}. \quad (13.25)$$

or, more compactly[16]

$$\hat{A}^{\mu}(x) = \int \frac{\mathrm{d}^3 p}{(2\pi)^{\frac{3}{2}}} \frac{1}{(2E_{\boldsymbol{p}})^{\frac{1}{2}}} \sum_{\lambda=1}^{3} \left(\epsilon_{\lambda}^{\mu}(p) \hat{a}_{\boldsymbol{p}\lambda} \mathrm{e}^{-\mathrm{i}p\cdot x} + \epsilon_{\lambda}^{\mu *}(p) \hat{a}_{\boldsymbol{p}\lambda}^{\dagger} \mathrm{e}^{\mathrm{i}p\cdot x} \right). \qquad (13.26)$$

Proca Equation Spin 1 particles

- In this case start w/ Maxwell's Equations

$$\partial_\mu \partial^\mu A^\nu - \partial^\nu \left(\partial_\mu A^\mu\right) = \frac{4\pi}{c} J^\nu$$

- Proca Equation

$$\partial_\mu F^{\mu\nu} + \left(\frac{mc}{\hbar}\right)^2 A^\nu = 0$$

This is known as the **Proca equation**. Taking the divergence gives us zero from the first term (since $\partial_\mu \partial_\nu F^{\mu\nu} = 0$) and we obtain from the second term $m^2 \partial_\nu A^\nu = 0$. Since $m^2 \neq 0$, we have the additional condition that the field obeys $\partial_\mu A^\mu = 0$. This allows us to write the Proca equations in another form

$$\left(\partial^2 + m^2\right) A^\nu = 0, \tag{13.19}$$

which looks a lot like the Klein–Gordon equation[10] for each of the four components of the field $A^\nu(x)$. The dispersion relation is therefore $E_p^2 = p^2 + m^2$ for each component.

For certain A gauge choices reduces to 4 Klein Gordons

Photon machinery

- Photon WF Solutions (solutions to Maxell d4Adx ∝ j

Plane-wave solutions with four-momentum $p = \left(\frac{E}{c}, \vec{p} \right)$ are of the form:

$$A^{\mu}(x) = a e^{-\frac{i}{\hbar} p \cdot x} \varepsilon^{\mu}(p)$$

Normalization factor (don't care here)

Polarization vector characterizing the photon spin-state.

$$\partial_{\mu}\partial^{\mu}A^{\nu} = 0 \quad \rightarrow \quad p_{\mu}p^{\mu} = 0 \quad \text{or} \quad E = |\vec{p}|c \quad \text{(correct for massless particle).}$$

The polarization vector ε^{μ} nominally has four components, but the Lorentz condition:

Feynman rules: incoming photon: $\varepsilon^{\mu}(p)$

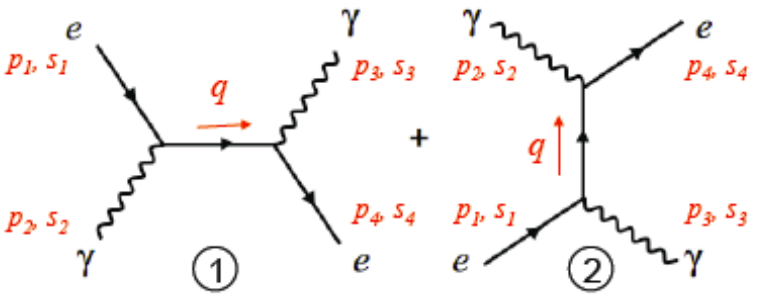
outgoing photon: $\varepsilon^{\mu^*}(p)$

$$\left\{ \varepsilon^{\mu}_{(s)} \text{ for } s = 1,2 \right\}$$

VERTEX : Still Follow Fermion Lines Bkwds

- Still follow advice to "follow backwards outgoing Fermion lines" always... (for Fermion-Fermion scat before, this is same as "(flux * flux)/propagator" for Halzen Martin simple Feynman (j ∝ p)
- in this case sandwiched between photon ϵ 's (and with Fermion propagator

Compton Scattering



For ①

write down contribution from fermion line:

Write down factors for two external photons (coupled to the correct vertex - e.g. with the correct μ or ν index)

Integrate over all internal momenta

$$\int \varepsilon_\mu^*(3)\left[\bar{u}(4)\, ig\gamma^\mu \frac{i(q+mc)}{q^2-m^2c^2} ig\gamma^\nu u(1)\right]\varepsilon_\nu(2)(2\pi)^4\delta^4(p_1+p_2-q)(2\pi)^4\delta^4(q-p_3-p_4)\frac{d^4q}{(2\pi)^4}$$

$$\mathcal{M}_1 = -\frac{g^2}{(p_1+p_2)^2-m^2c^2}\left[\bar{u}(4)\boldsymbol{\varepsilon}^*(3)(p_1+p_2+mc)\boldsymbol{\varepsilon}(2)u(1)\right]$$

Compton Spin

$$\mathcal{M}_2 = \frac{g_e^2}{(p_1+p_2)^2 - m^2c^2}\left[\bar{u}(4)\not{\varepsilon}(3)^*(\not{p}_1 + \not{p}_2 + mc)\not{\varepsilon}(2)u(1)\right]$$

- Sum over all spin states of photon
- after all the trace theorems it looks like this because we can factor out both polarization vectors

$$\sum_{\gamma spin} \varepsilon^*(3)_\mu \varepsilon(2)_\nu \, T^{\mu\nu}$$

Completeness for photons two times

$$\sum'_{\gamma spin} \varepsilon^*(3)_\mu \varepsilon(2)_\nu \sum_{\gamma' spin} \varepsilon_\kappa(3) \varepsilon^*_{\alpha'}(2) \, T^{\mu\nu\kappa\alpha'}$$

"$Q_{\mu\kappa} Q_{\nu\alpha}$"

$$T = \mathrm{Tr}\left[\gamma^\mu(\not{p}_1 + \not{p}_2)\gamma^\nu \not{p}_1 \gamma^\kappa (\not{p}_1 + \not{p}_2)\gamma^\alpha \not{p}_4\right]$$

This is the answer to the M_2 part of Griff Prob. 7.32 – also shown in slides is H&M version–compare, I said they look like they're ~the same (to a~sign)

For s diagram

$$\mathcal{M}_1 = -\frac{g^2}{(p_1+p_2)^2 - m^2c^2}\left[\bar{u}(4)\varepsilon^*(3)(\not{p}_1 + \not{p}_2 + mc)\varepsilon(2)u(1)\right]$$

keep stars

s

"Γ" $= \not{\varepsilon}_3^*(\not{p}_1+\not{p}_2)\not{\varepsilon}_2$

$$|\mathcal{M}_1^2| = \left\{\frac{g^4}{s^2}\right\}[\bar{u}_4\Gamma u_1][\bar{u}_4\Gamma u_1]^*$$

ε'

$$\bar{u}_4\Gamma u_1\,\bar{u}_1\bar{\Gamma}u_4$$

$$\rightarrow \bar{u}_4\Gamma\not{p}_1\bar{\Gamma}u_4 \Rightarrow \mathrm{Tr}(\Gamma\not{p}_1\bar{\Gamma}\not{p}_4)$$

- Can just use this relation:

$$\bar{\Gamma} = \not{\varepsilon}_2^*(\not{p}_1+\not{p}_2)\not{\varepsilon}_3$$

For s diagram

$$\mathcal{M}_1 = -\frac{g^2}{(p_1+p_2)^2 - m^2c^2}\left[\bar{u}(4)\varepsilon^*(3)(\not{p}_1 + \not{p}_2 + mc)\varepsilon(2)u(1)\right]$$

keep stars

"Γ" $= \not{\varepsilon}_3^*(\not{p}_1 + \not{p}_2)\not{\varepsilon}_2$

$$\mathcal{M}_1^2 = \frac{g^2}{s^2}\left[\bar{u}_4 \Gamma u_1\right]\left[\bar{u}_4 \Gamma u_1\right]^*$$

$$\left(\bar{u}_4 \Gamma u_1\right)\left(\bar{u}_1 \bar{\Gamma} u_4\right)$$

$$\bar{u}_4 \Gamma \not{p}_1 \bar{\Gamma} u_4 \Rightarrow Tr\left(\Gamma \not{p}_1 \bar{\Gamma} \not{p}_4\right)$$

- Can just use this relation:

$$\bar{\Gamma} = \not{\varepsilon}_2^*(\not{p}_1 + \not{p}_2)\not{\varepsilon}_3$$

Completeness for photons - Griff

- One of these points was actuallu made later in class

Feynman rules: incoming photon: $\varepsilon^{\mu}(p)$

outgoing photon: $\varepsilon^{\mu *}(p)$

$\left\{\varepsilon^{\mu}_{(s)} \text{ for } s=1,2\right\}$

Here ε^{μ} is the polarization vector which satisfies the following properties:

$$\varepsilon^{\mu} p_{\mu} = 0 \qquad \varepsilon^{\mu *}_{(1)} \varepsilon_{(2)\mu} = 0 \ \text{(orthogonal)} \qquad \varepsilon^{\mu *} \varepsilon_{\mu} = 1 \ \text{(normalized)}$$

In the Coloumb gauge: $\varepsilon^{0} = 0, \ \vec{\varepsilon} \cdot \vec{p} = 0 \ \ (\text{since } \varepsilon^{\mu} p_{\mu} = 0)$

If p is chosen to be along the z direction, one choice is: $\varepsilon_1 = (1,0,0) \ \ \varepsilon_2 = (0,1,0)$

DON''T USE THIS ONE FOR PROBLEMS

Different gauge than most text books use

(see next slide)

The completeness relation is: $\sum_{s=1,2} \left(\varepsilon_{(s)}\right)_i \left(\varepsilon_{(s)}\right)_j = \delta_{ij} - \hat{p}_i \hat{p}_j$.

For example, in the case where p is along z, $\vec{p} = |\vec{p}|\hat{z}$

$$\hat{p}_i \hat{p}_j = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{so} \quad \delta_{ij} - \hat{p}_i \hat{p}_j = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Halzen Martin Compton – alt Completeness for Photons

- Instead use $\Sigma\epsilon\epsilon = -g_{\mu\nu}$ instead of = $\delta ij - pipj$
- ~All other particle physics books do

The two invariant amplitudes, (6.103) and (6.104), are

$$\mathcal{M}_1 = \epsilon'^{*}_{\nu}\epsilon_{\mu}e^2\,\bar{u}(p')\,\gamma^{\nu}(\not{p}+\not{k})\gamma^{\mu}\,u(p)/s,$$
$$\mathcal{M}_2 = \epsilon'^{*}_{\nu}\epsilon_{\mu}e^2\,\bar{u}(p')\,\gamma^{\mu}(\not{p}-\not{k}')\gamma^{\nu}\,u(p)/u. \qquad (6.110)$$

To obtain the unpolarized cross section, we must average/sum $|\mathcal{M}_1 + \mathcal{M}_2|^2$ over the initial/final electron and photon spins. Fortunately, this is not as difficult as it first appears. For physical photons, (6.98) applies, and we can make the replacement

$$\sum_T \epsilon^{T*}_{\mu}\epsilon^{T}_{\mu'} \to -g_{\mu\mu'} \qquad (6.111)$$

where T denotes transverse. We have a similar completeness relation for the outgoing photon states, ϵ'. Thus, for example,

$$\overline{|\mathcal{M}_1|^2} = \frac{e^4}{4s^2}\sum_{s,s'}\left(\bar{u}^{(s')}\gamma^{\nu}(\not{p}+\not{k})\gamma^{\mu}u^{(s)}\right)\left(\bar{u}^{(s)}\gamma_{\mu}(\not{p}+\not{k})\gamma_{\nu}u^{(s')}\right).$$

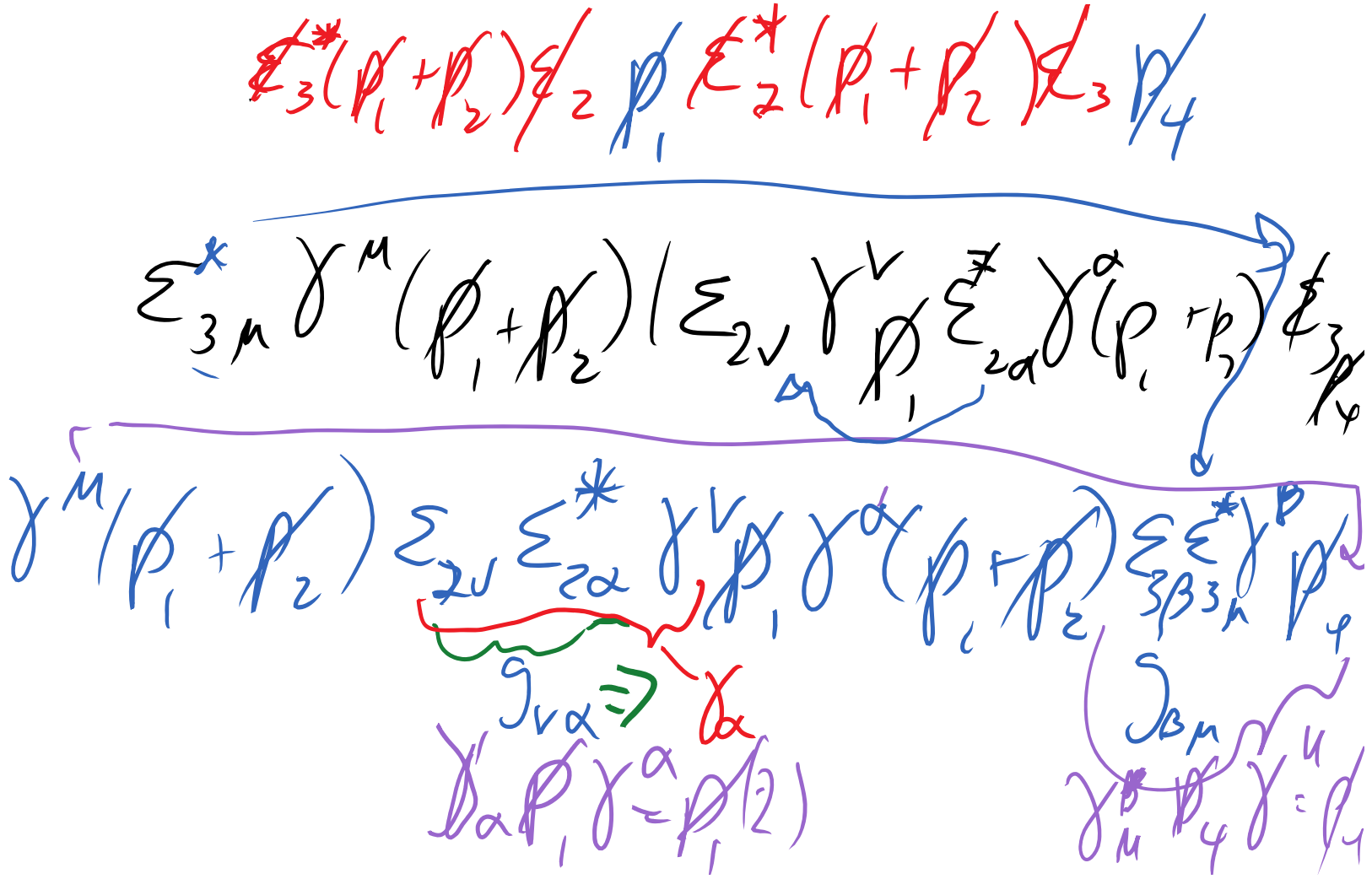
The factor $\frac{1}{4}$ is due to averaging over the initial electron and photon spins. The spinor completeness relation, (5.47), allows the sum over $u\bar{u}$ states to be performed (just as we described for $e^-\mu^-$ scattering in Section 6.3), and we find

$$\overline{|\mathcal{M}_1|^2} = \frac{e^4}{4s^2}\mathrm{Tr}\left(\not{p}'\gamma^{\nu}(\not{p}+\not{k})\gamma^{\mu}\not{p}\gamma_{\mu}(\not{p}+\not{k})\gamma_{\nu}\right)$$

6.98 is exactly the same as 6.111

see next slide for continuation of this line

- Use guv thing for inside trace



- Use guv thing for inside trace

$$\not{\epsilon}_3^*(\not{p}_1+\not{p}_2)\not{\epsilon}_2\not{p}_1\not{\epsilon}_2^*(\not{p}_1+\not{p}_2)\not{\epsilon}_3\not{p}_4$$

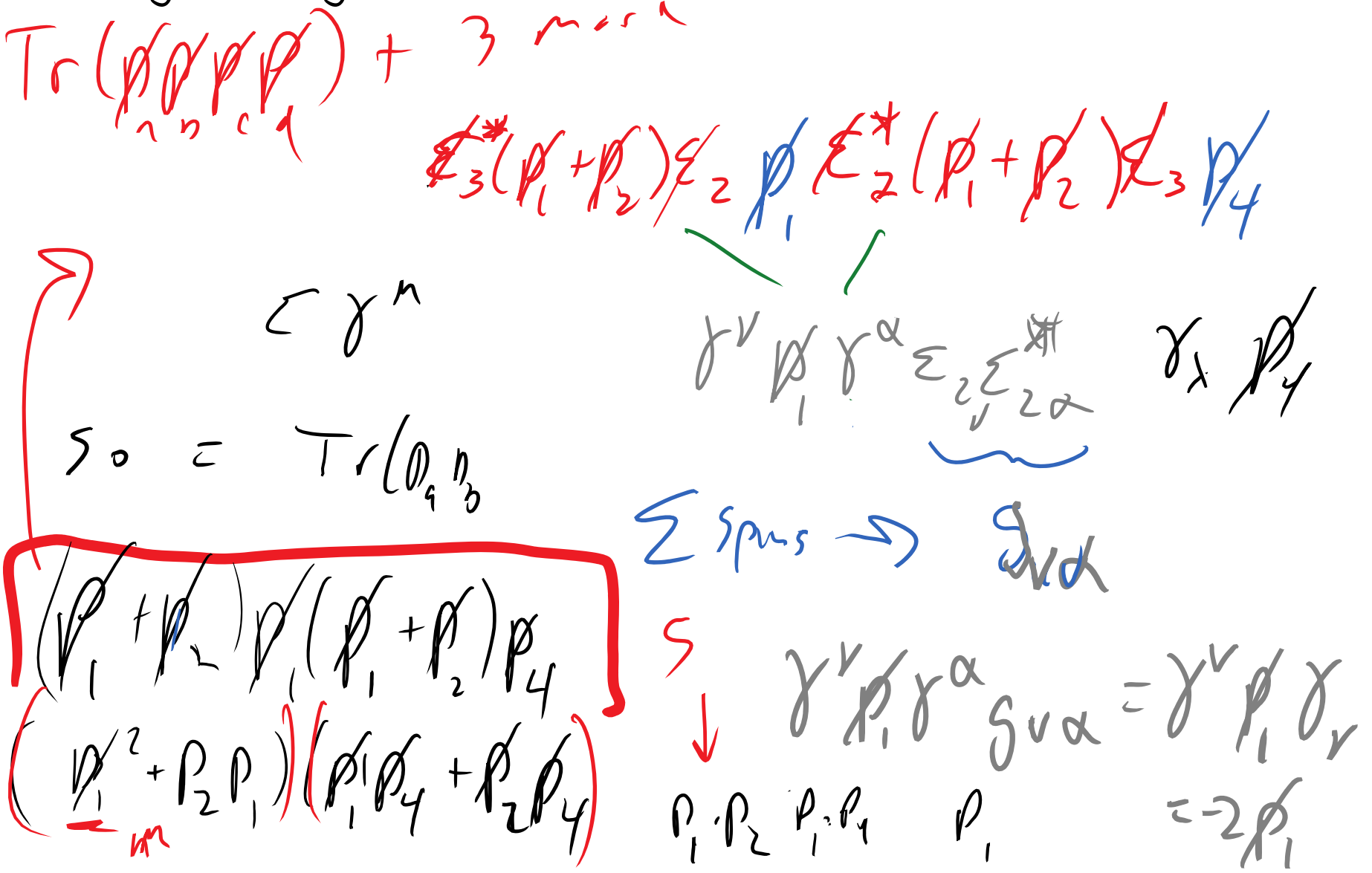
$$\gamma^\nu \not{p}_1 \gamma^\alpha \epsilon_{2\nu}\epsilon_{2\alpha}^*$$

$$\sum s_{pols} \rightarrow g_{\nu\alpha}$$

$$\gamma^\nu \not{p}_1 \gamma^\alpha g_{\nu\alpha} = \gamma^\nu \not{p}_1 \gamma_\nu$$

$$= -2\not{p}_1$$

- Use guv thing for inside trace



Halzen Martin Compton – alt Completeness for Photons

- For actually doing the photon ϵ calcs with trace/sum over spins – H&M suggests something which Griff doesn't mention…in Griff solutions for Compton also perhaps even in disagreement HM: use $\Sigma\epsilon\epsilon = -g_{\mu\nu}$ instead of = $\delta ij - pipj$

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$$\overline{|\mathcal{M}_1|^2} = \frac{e^4}{4s^2}\mathrm{Tr}\left(\underbrace{\not{p}'\gamma^{\nu}}_{2\not{p}'}(\not{p}+\not{k})\underbrace{\gamma^{\mu}\not{p}\gamma_{\mu}}_{-2\not{p}}(\not{p}+\not{k})\gamma_{\nu}\right)$$

6.98 is exactly the same as 6.111

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$$\overline{|\mathcal{M}_1|^2} = \frac{e^4}{4s^2}\mathrm{Tr}\,(\underbrace{\not{p}'\gamma^\nu}_{-2\not{p}'}(\not{p}+\not{k})\underbrace{\gamma^\mu \not{p}\gamma_\mu}_{-2\not{p}}(\not{p}+\not{k})\gamma_\nu)$$

$$= \frac{e^4}{s^2}\mathrm{Tr}(\not{p}'\not{k}\not{p}\not{k})$$

$$= \frac{4e^4}{s^2}2(p'\cdot k)(p\cdot k)$$

$$= 2e^4\left(-\frac{u}{s}\right), \qquad (6.112)$$

e use of (6.24) and (6.22). Similarly, we obtain

$$\overline{|\mathcal{M}_2|^2} = 2e^4\left(-\frac{s}{u}\right),$$

$$\overline{\mathcal{M}_1\mathcal{M}_2^*} = 0.$$

$(p'\not{p}+p'\not{k})(\not{p}+\not{k})\not{p}'(\not{p}+\not{k})\not{p}(\not{p}+\not{k})$

Compton Halzen&Mart

- Compton scattering esp ending part:

$$\overline{|\mathcal{M}_1|^2} = \frac{e^4}{4s^2}\mathrm{Tr}\,(\underbrace{\not p'\gamma^\nu}_{-2\not p'}(\not p + \not k)\underbrace{\gamma^\mu \not p \gamma_\mu}_{-2\not p}(\not p + \not k)\gamma_\nu)$$

$$= \frac{e^4}{s^2}\mathrm{Tr}(\not p'\not k\not p\not k)$$

$$= \frac{4e^4}{s^2}2(p'\cdot k)(p\cdot k)$$

$$= 2e^4\left(-\frac{u}{s}\right),$$

ıave made use of (6.24) and (6.22). Similarly, we obtain

$$\overline{|\mathcal{M}_2|^2} = 2e^4\left(-\frac{s}{u}\right),$$

$$\overline{\mathcal{M}_1\mathcal{M}_2^*} = 0.$$

Thus, the spin-averaged Compton amplitude is

$$\overline{|\mathcal{M}|^2} = \overline{|\mathcal{M}_1 + \mathcal{M}_2|^2} = 2e^4\left(-\frac{u}{s} - \frac{s}{u}\right).$$

EXERCISE 6.19 Repeat the above calculation for an photon of mass $k^2 \equiv -Q^2$. Continue to use (6.111). Show γe^- (where γ^* denotes a virtual photon),

$$\overline{|\mathcal{M}|^2} = 2e^4\left(-\frac{u}{s} - \frac{s}{u} + \frac{2Q^2t}{su}\right).$$

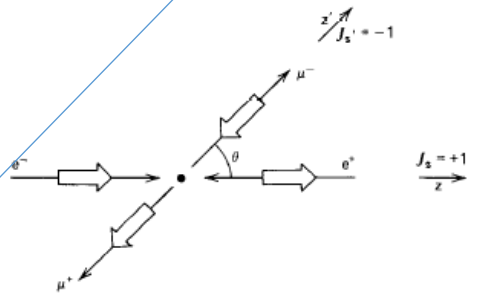


Fig. 6.9 The process $e^-e^+ \to \mu^-\mu^+$ in a particular helicity configuration: $\mathcal{M}(e_R^-e_L^+ \to \mu_L^-\mu_R^+) \sim d^1_{-11}(\theta)$.

This answer is definitely right it agrees with other texts and can be justified another way by Helicity conservation which I also wanted to mention because that way uses something from advanced QM: d(θ) functions… --evidence for the $-g^{\mu\nu}$ completeness relation?

$$d^1_{11}(\theta) = d^1_{-1-1}(\theta) = \tfrac{1}{2}(1 + \cos\theta) = -\frac{u}{s}$$

One can use ϵ vectors...

- If one goes into the "2x2" form as shown by Griff for the e+e-
  - In his example, he is only choosing one spin state (and "circular for the first time ;<...this could still be done for sum over spin states– another option then using "tensor products"

$$\not{\epsilon} = -\epsilon \cdot \gamma = -\begin{pmatrix} 0 & \sigma \cdot \epsilon \\ -\sigma \cdot \epsilon & 0 \end{pmatrix}$$

therefore

$$\not{\epsilon}_3 \not{\epsilon}_4 = \begin{pmatrix} 0 & \sigma \cdot \epsilon_3 \\ -\sigma \cdot \epsilon_3 & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma \cdot \epsilon_4 \\ -\sigma \cdot \epsilon_4 & 0 \end{pmatrix}$$

$$= -\begin{pmatrix} (\sigma \cdot \epsilon_3)(\sigma \cdot \epsilon_4) & 0 \\ 0 & (\sigma \cdot \epsilon_3)(\sigma \cdot \epsilon_4) \end{pmatrix}$$

In Chapter 4 (Problem 4.20) we encountered th

$$(\sigma \cdot \mathbf{a})(\sigma \cdot \mathbf{b}) = \mathbf{a} \cdot \mathbf{b} + i\sigma \cdot (\mathbf{a} \times \mathbf{b})$$

$$\mathcal{M} = \frac{g_e^2}{mc} \bar{v}(2) \left[(\epsilon_3 \cdot \epsilon_4)\gamma^0 + i(\epsilon_3 \times \epsilon_4) \cdot \Sigma \gamma^3 \right] u(1)$$

Epsilon usage

- at high energy seems unlikely we would want circular polarization... thinking of helicity. photons produced from electron bremsstrahlung but summing over these must be the same.

$$\sum_{\gamma\,spin} \varepsilon^{*}(3)_\mu \, \varepsilon(2)_\nu \, T^{\mu\nu}$$

scattered photon is NOT in z-direction but w/ ang θ

$$\hat{p}_\gamma^{f} = \begin{pmatrix} \sin\theta \\ 0 \\ \cos\theta \end{pmatrix} \Rightarrow \begin{pmatrix} 0 \\ -\cos\theta \\ 0 \\ \sin\theta \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

We always assume $\mathbf{p}_i$'s in z direction (perp that)

$$\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

can choose xz plane (phi=0; not used tho) to contain pf's

$\cos^2$

$$\delta_{ij} - \hat{p}_i \hat{p}_j = \begin{pmatrix} 1-\sin^2\theta & 0 & -\cos\theta \\ 0 & 1 & 0 \\ -\cos\theta & 0 & 1-\cos^2\theta \end{pmatrix}$$

Four non-zero terms

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

two non-zero terms

Epsilon usage

- at high energy seems unlikely we would want circular polarization… thinking of helicity. photons produced from electron bremsstrahlung but summing over these must be the same.

not dot products show below

$$\sum_{\gamma\,spin} \varepsilon^{*}(3)_\mu \, \varepsilon(2)_\nu \, T^{\mu\nu}$$

scattered photon is NOT in z-direction but w/ ang θ

$$\hat{p}_\gamma^{f} = \begin{pmatrix} \sin\theta \\ 0 \\ \cos\theta \end{pmatrix} \Rightarrow \begin{pmatrix} 0 \\ -\cos\theta \\ 0 \\ \sin\theta \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$\cos^2$

can choose xz plane (phi=0; not used tho) to contain pf's

$$\delta_{ij} - \hat{p}_i \hat{p}_j = \begin{pmatrix} 1-\sin^2\theta & 0 & -c \\ 0 & 1 & 0 \\ -c & 0 & 1-c^2 \end{pmatrix}$$

Four non-zero terms

We always assume p_i's in z direction (perp that)

$$\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

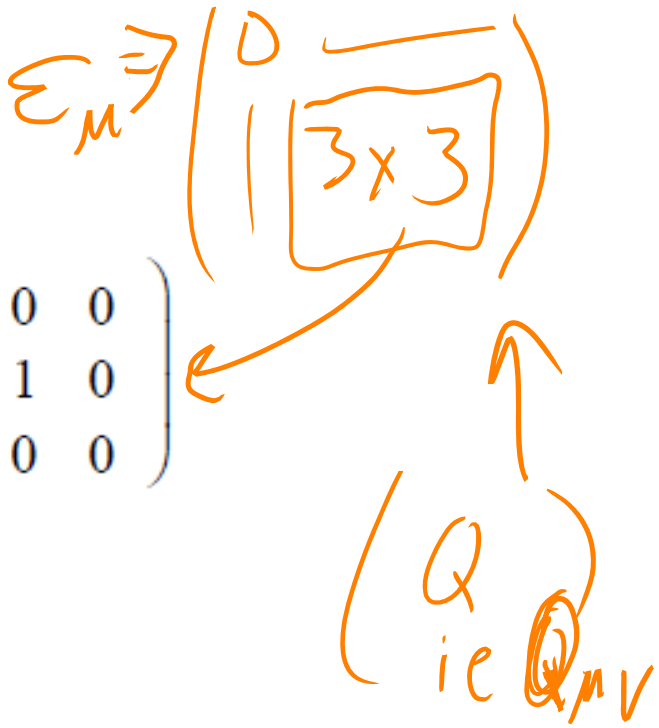
two non-zero terms

Focus in on "Completeness Matrices"

- Note given relations are 3x3 0-comp = 0

$$\delta_{ij} - \hat{p}_i \hat{p}_j = \begin{pmatrix} 1-\sin^2\theta\cos^2\phi & -\sin^2\theta\sin\phi\cos\phi & -s c \\ -\sin^2\theta\sin\phi\cos\phi & 1-\sin^2\theta\sin^2\phi & 0 \\ -s c & 0 & 1-c^2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$



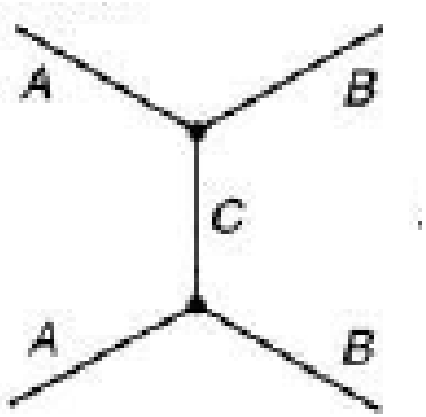
Nonzero combos for (note) μνκα

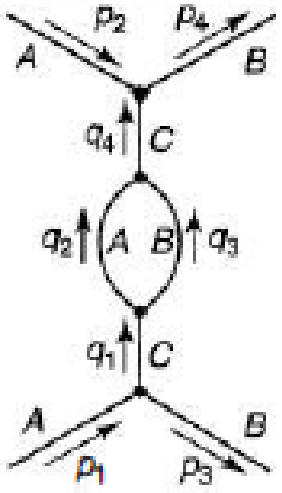
1111, 1212, 1131,,1232,3111,3212,3131,3232 w/ factors
c^2, c^2, -sc -sc -sc -sc ,s^2 s^2

It would appear that there could be some additional angular dependence from these factors-- but it likely cancels? Ahh... look for web reference!
Found it, yes it does add extra ang dependence terms!

Renorm discussion – δ fn Counting

- Feyman Rules add loops

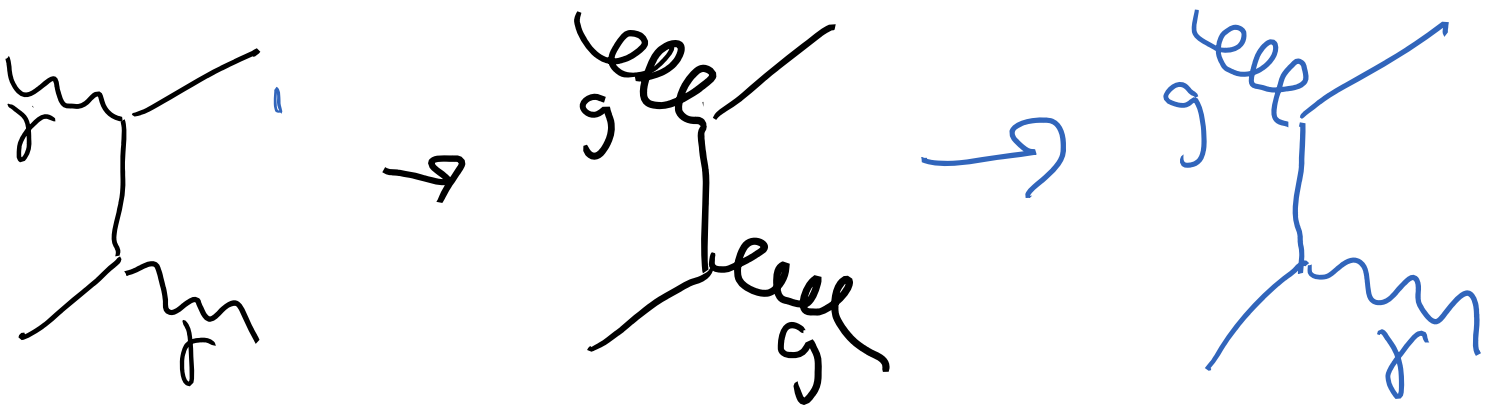




- AGAIN FOR QED : more integrals added then delta functions
- Higher order → need "real" integration
- Leads to running coupling constant in QED (coupling increases w/ higher energy→ opposite to QCD

Compton → QCD

- Very nice web reference
  - https://sites.ualberta.ca/~gingrich/courses/phys512/node102.html
  - Another Canadian site!!!
- COMPTON VERY IMPORTANT FOR QCD
- **It's worth exploring Compton Scattering in detail** because its quite important in QCD because gluon's are so prevalent in hadrons --
  - **GLUON Compton scattering**
  - M calcs exactly the same → only diff is extra "color" factor



From Compton Ref

Photon-Electron Scattering

$$\boxed{\frac{d\bar{\sigma}}{d\Omega}(\lambda,\lambda') = \frac{\alpha^2}{4m^2}\left(\frac{\omega'}{\omega}\right)^2\left[\frac{\omega'}{\omega}+\frac{\omega}{\omega'}+4(\varepsilon'\cdot\varepsilon)^2-2\right]}, \qquad (7.245)$$

the Klein-Nishina formula for Compton scattering.

$$\hat{k}' = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta).$$

$$\vec{\varepsilon}_{(1)} = (1,0,0), \qquad \varepsilon'_{(1)} = (\sin\phi, -\cos\phi, 0),$$

$$\vec{\varepsilon}_{(2)} = (0,1,0), \qquad \varepsilon'_{(2)} = (\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta).$$

1 and the cross-section reduces to the

$$\boxed{\frac{d\bar{\sigma}}{d\Omega} = \frac{\alpha^2}{2m^2}\left(\frac{\omega'}{\omega}\right)^2\left(\frac{\omega'}{\omega}+\frac{\omega}{\omega'}-\sin^2\theta\right)}.$$

The low-energy for forward-scattering limit (classical limit) now becomes

$$\boxed{\left(\frac{d\bar{\sigma}}{d\Omega}\right)_{\text{class}} = \frac{r_0^2}{2}\left(1+\cos^2\theta\right)}$$

Halzen Martin Compton – alt Completeness for Photons

- For actually doing the photon ϵ calcs with trace/sum over spins – H&M suggests something which Griff doesn't mention…in Griff solutions for Compton also perhaps even in disagreement HM: use $\Sigma\epsilon\epsilon = -g_{\mu\nu}$ instead of = $\delta ij - pipj$

The two invariant amplitudes, (6.103) and (6.104), are

$$\mathcal{M}_1 = \epsilon'^{*}_{\nu}\epsilon_{\mu}e^2\,\bar{u}(p')\,\gamma^{\nu}(\not{p}+\not{k})\gamma^{\mu}\,u(p)/s,$$
$$\mathcal{M}_2 = \epsilon'^{*}_{\nu}\epsilon_{\mu}e^2\,\bar{u}(p')\,\gamma^{\mu}(\not{p}-\not{k}')\gamma^{\nu}\,u(p)/u. \qquad (6.110)$$

To obtain the unpolarized cross section, we must average/sum $|\mathcal{M}_1 + \mathcal{M}_2|^2$ over the initial/final electron and photon spins. Fortunately, this is not as difficult as it first appears. For physical photons, (6.98) applies, and we can make the replacement

$$\sum_T \epsilon^{T*}_{\mu}\epsilon^{T}_{\mu'} \to -g_{\mu\mu'} \qquad (6.111)$$

where T denotes transverse. We have a similar completeness relation for the outgoing photon states, ϵ'. Thus, for example,

$$\overline{|\mathcal{M}_1|^2} = \frac{e^4}{4s^2}\sum_{s,s'}\left(\bar{u}^{(s')}\gamma^{\nu}(\not{p}+\not{k})\gamma^{\mu}u^{(s)}\right)\left(\bar{u}^{(s)}\gamma_{\mu}(\not{p}+\not{k})\gamma_{\nu}u^{(s')}\right).$$

The factor $\frac{1}{4}$ is due to averaging over the initial electron and photon spins. The spinor completeness relation, (5.47), allows the sum over $u\bar{u}$ states to be performed (just as we described for $e^-\mu^-$ scattering in Section 6.3), and we find

$$\overline{|\mathcal{M}_1|^2} = \frac{e^4}{4s^2}\mathrm{Tr}\left(\underbrace{\not{p}'\gamma^{\nu}}_{2\not{p}'}(\not{p}+\not{k})\underbrace{\gamma^{\mu}\not{p}\gamma_{\mu}}_{-2\not{p}}(\not{p}+\not{k})\gamma_{\nu}\right)$$

6.98 is exactly the same as 6.111

see next slide for continuation of this line

Compton H&M

- Compton scattering esp ending part:

$$\overline{|\mathcal{M}_1|^2} = \frac{e^4}{4s^2}\mathrm{Tr}\,(\underbrace{\not p'\gamma^\nu}_{-2\not p'}(\not p + \not k)\underbrace{\gamma^\mu \not p \gamma_\mu}_{-2\not p}(\not p + \not k)\gamma_\nu)$$

$$= \frac{e^4}{s^2}\mathrm{Tr}(\not p'\not k\not p\not k)$$

$$= \frac{4e^4}{s^2}2(p'\cdot k)(p\cdot k)$$

$$= 2e^4\left(-\frac{u}{s}\right),$$

ıave made use of (6.24) and (6.22). Similarly, we obtain

$$\overline{|\mathcal{M}_2|^2} = 2e^4\left(-\frac{s}{u}\right),$$

$$\overline{\mathcal{M}_1\mathcal{M}_2^*} = 0.$$

Thus, the spin-averaged Compton amplitude is

$$\overline{|\mathcal{M}|^2} = \overline{|\mathcal{M}_1 + \mathcal{M}_2|^2} = 2e^4\left(-\frac{u}{s} - \frac{s}{u}\right).$$

EXERCISE 6.19 Repeat the above calculation for an photon of mass $k^2 \equiv -Q^2$. Continue to use (6.111). Show γe^- (where γ^* denotes a virtual photon),

$$\overline{|\mathcal{M}|^2} = 2e^4\left(-\frac{u}{s} - \frac{s}{u} + \frac{2Q^2t}{su}\right).$$

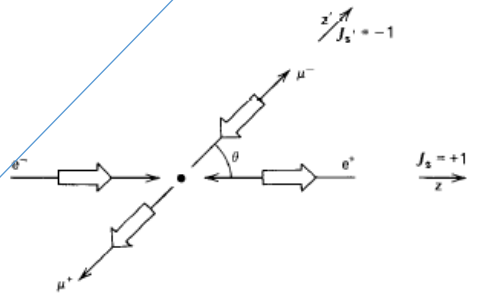


Fig. 6.9 The process $e^-e^+ \to \mu^-\mu^+$ in a particular helicity configuration: $\mathcal{M}(e_R^-e_L^+ \to \mu_L^-\mu_R^+) \sim d^1_{-11}(\theta)$.

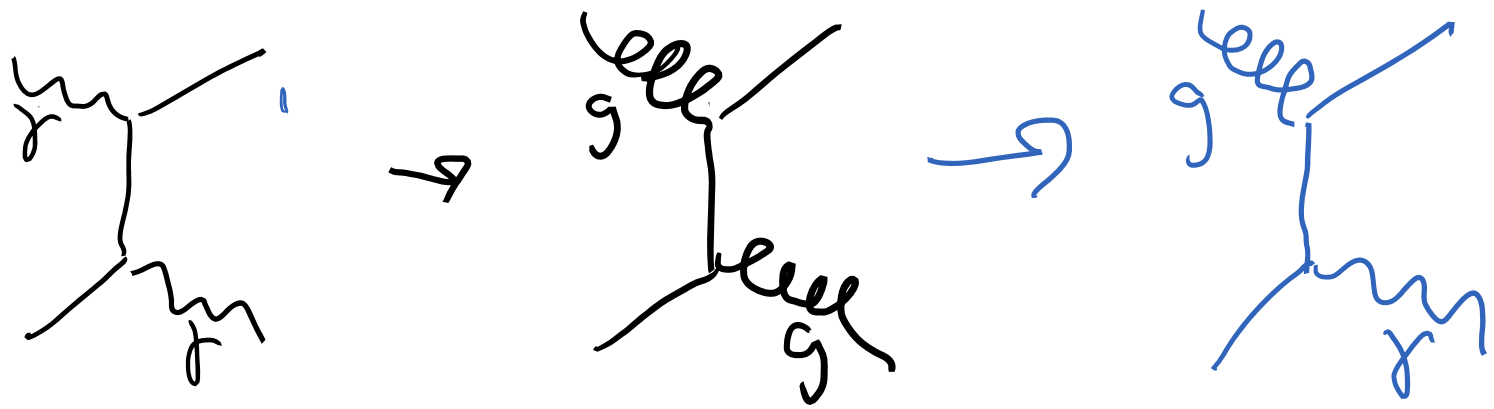
This answer is definitely right it agrees with other texts and can be justified another way by Helicity conservation which I also wanted to mention because that way uses something from advanced QM: d(θ) functions… --evidence for the $-g^{\mu\nu}$ completeness relation?

$$d^1_{11}(\theta) = d^1_{-1-1}(\theta) = \tfrac{1}{2}(1 + \cos\theta) = -\frac{u}{s}$$

Compton Scattering IMPORTANT →QCD

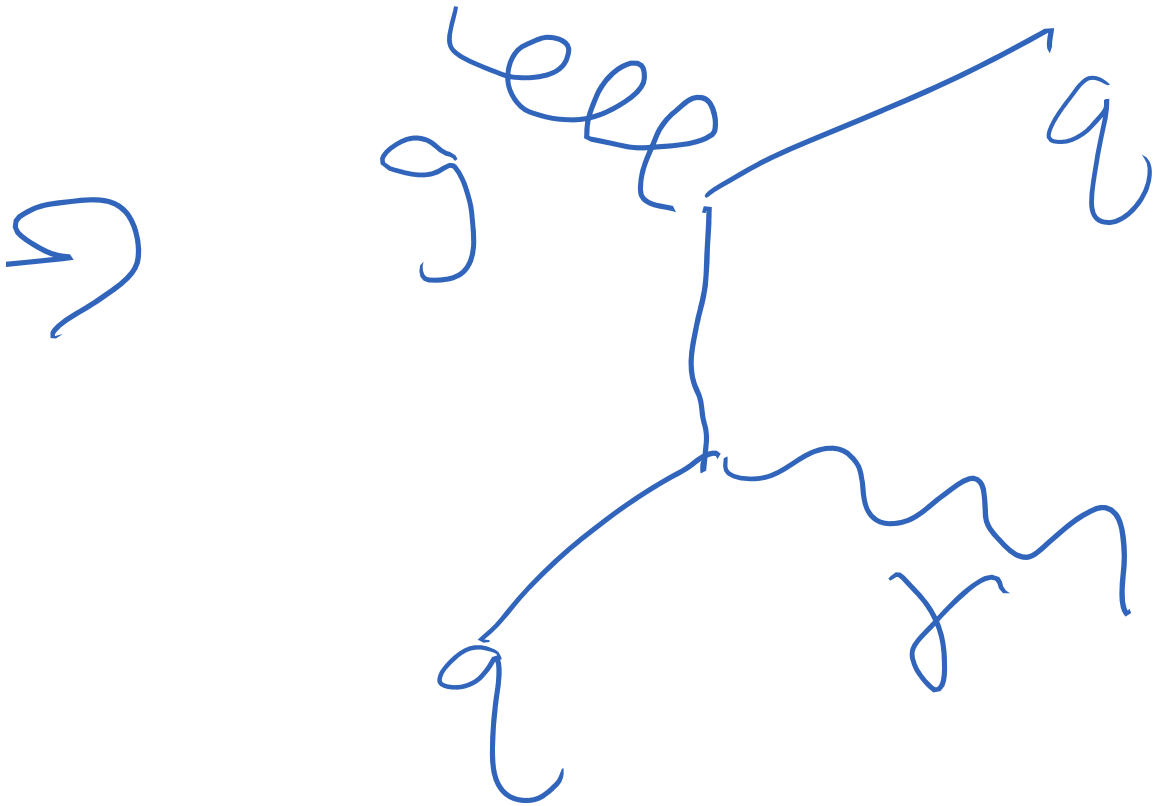
- Very nice web reference
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  - Another Canadian site!!!

- **It's worth exploring Compton Scattering in detail** because its quite important in QCD because gluon's are so prevalent in hadrons --
  - **GLUON Compton scattering**
  - M calcs exactly the same → only diff is extra "color" factor



Feynman Diagrams

- Also notice many first calcs, are a mix of QED and QCD



Any outgoing q or g is a JET! JETS CAN DEMONSTRATE basic properties of quarks!!! color/spin!

Chapter 8 QCD/Pre QCD

Consider the high energy limit, $E >> Mc^2$. In that case we obtain $\sigma\left(e^+e^- \to f\bar{f}\right) = \frac{\pi}{3}\left(\frac{\hbar c Q_f \alpha}{E}\right)^2$.

Now consider the ratio of the cross-section for the production of hadrons in e^+e^- collisions relative to the cross-section for producing a muon pair:

$$R(E) = \frac{\sigma\left(e^+e^- \to hadrons\right)}{\sigma\left(e^+e^- \to \mu^+\mu^-\right)}$$

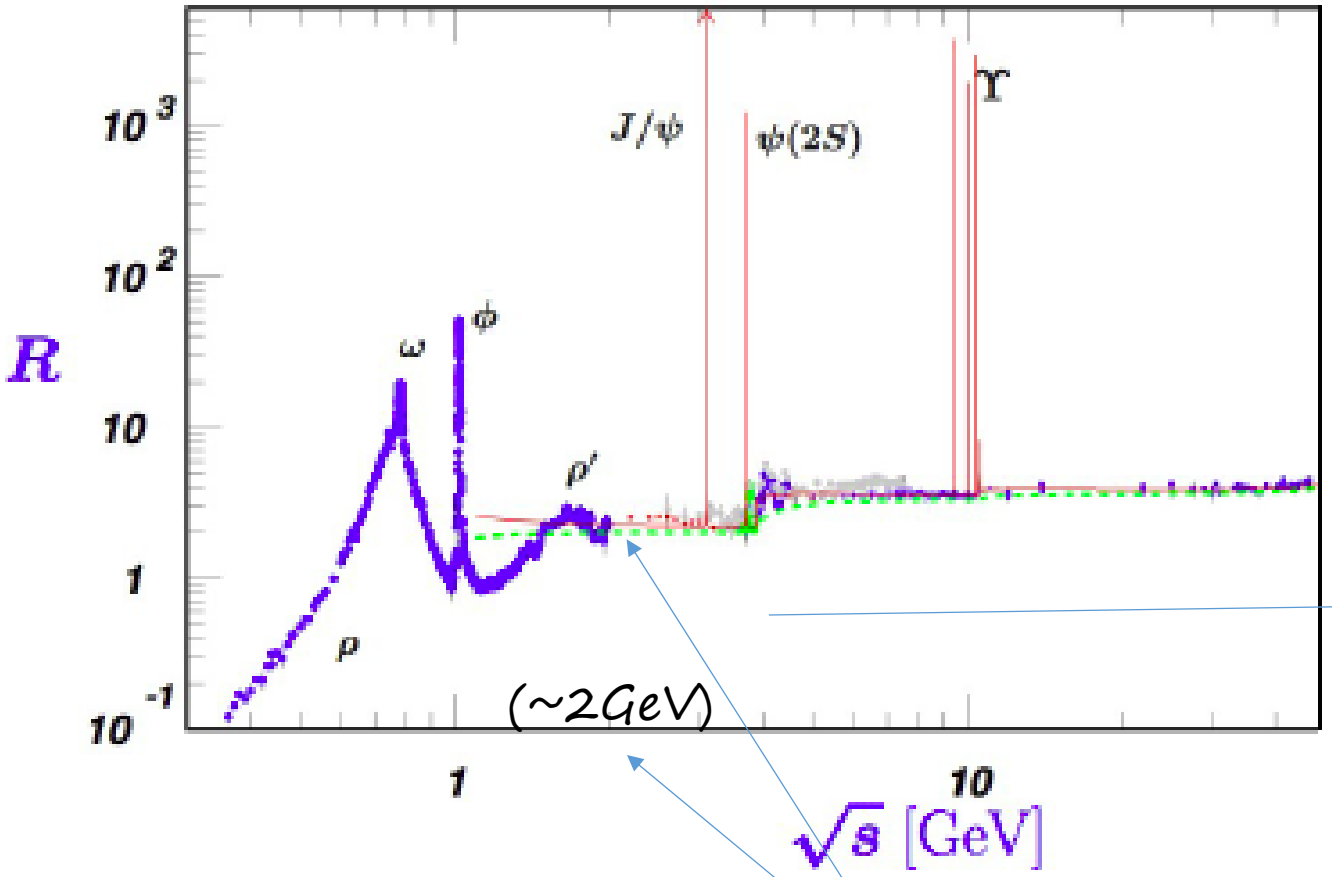
This is a function of energy because the quarks that are kinematically accessible (for the process in the numerator) depends on the energy. As the energy rises we pass various thresholds for the production of new quark-antiquark pairs in the final state.

$$R(E) = 3\sum_{f=u,d,s,c,b,t} Q_f^2$$

Here we need to sum over all kinematically accessible quark-antiquark final states. Where does factor of three come from?

SHOULD BE e+e- → (qqbar) JETS for which this is valid
You see it it only kicks in at the mimimum jet energy (> ~2 GeV)
(below that we don't consider it a perturbative jet)

Famous plot—newer vers than Griff



w/o color D.O.F.

- Statements only good starting here.
- Steps above that are from new flavors (3, then 4 (*4/3))…but these jumps are small to the overall factor of three compared to color

QED→QCD

- QCD Feyman Rules "include" QED's but add new parts:

Gluon-Gluon Interactions

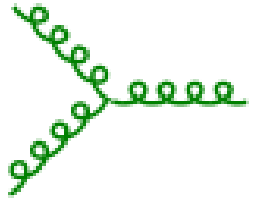
★ In QED the photon does not carry the charge of the EM interaction (photons are electrically neutral)

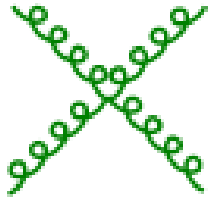
★ In contrast, in QCD the gluons do carry colour charge

⇨ Gluon Self-Interactions

★ Two new vertices (no QED analogues)

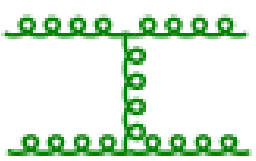
triple-gluon vertex

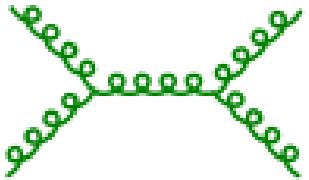




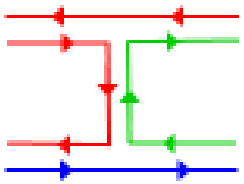
quartic-gluon vertex

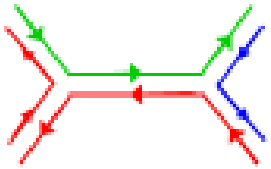
★ In addition to quark-quark scattering, therefore can have gluon-gluon scattering





e.g. possible way of arranging the colour flow



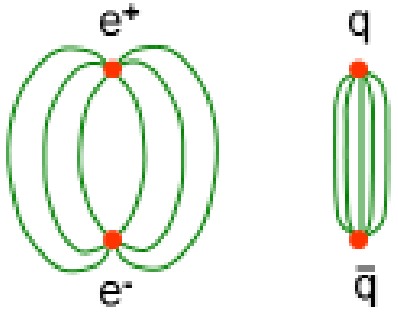


Gluon self-Interactions and Confinement

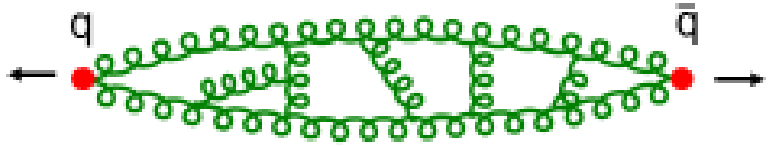
★ Gluon self-interactions are believed to give rise to colour confinement

★ Qualitative picture:

- Compare QED with QCD
- In QCD "gluon self-interactions squeeze lines of force into a flux tube"



★ What happens when try to separate two coloured objects e.g. $q\bar{q}$



- Form a flux tube of interacting gluons of approximately constant energy density $\sim 1\,\mathrm{GeV/fm}$

$$\Rightarrow \quad V(r) \sim \lambda r$$

- Require infinite energy to separate coloured objects to infinity
- Coloured quarks and gluons are always confined within colourless states
- In this way QCD provides a plausible explanation of confinement – but not yet proven (although there has been recent progress with Lattice QCD)

Hadronisation and Jets

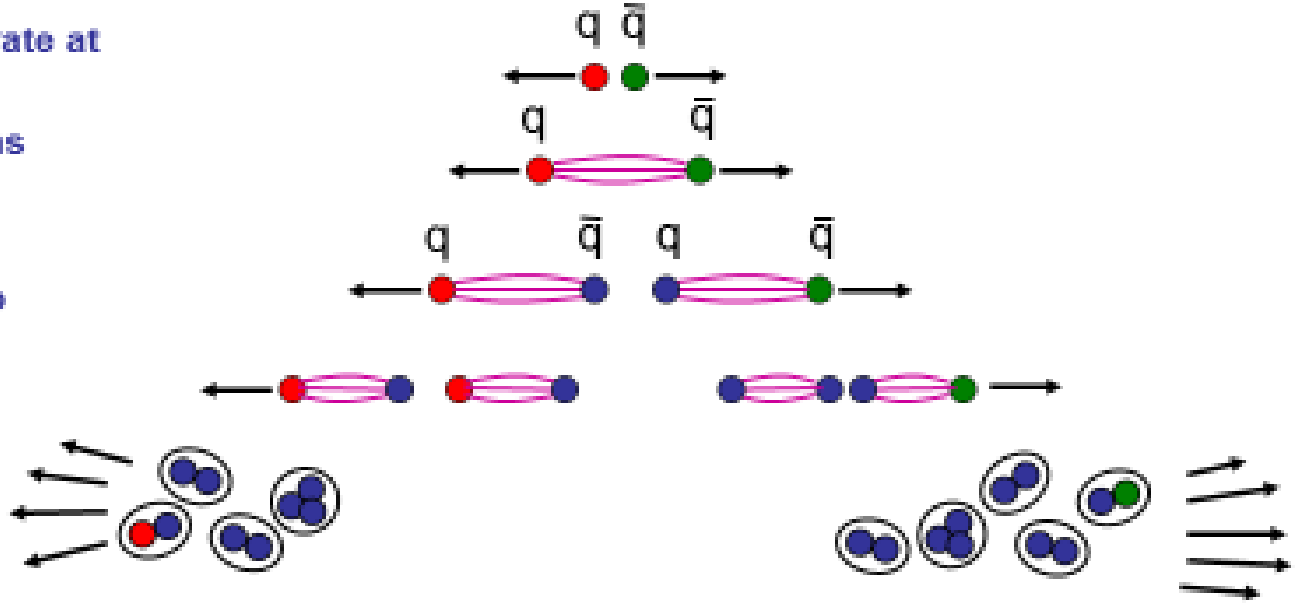
★ Consider a quark and anti-quark produced in electron positron annihilation

i) Initially Quarks separate at high velocity

ii) Colour flux tube forms between quarks

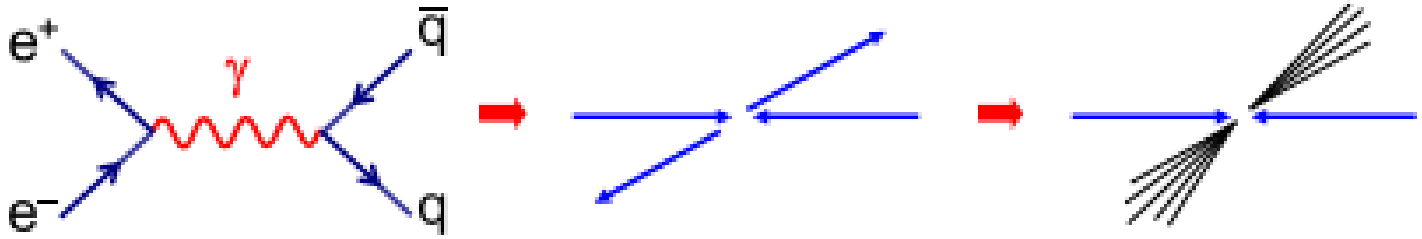
iii) Energy stored in the flux tube sufficient to produce $q\bar{q}$ pairs

iv) Process continues until quarks pair up into jets of colourless hadrons



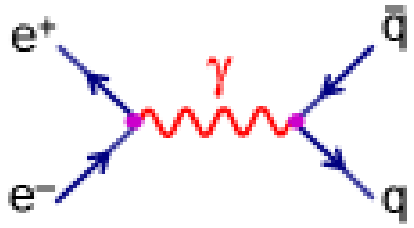
★ This process is called **hadronisation**. It is not (yet) calculable.

★ The main consequence is that at collider experiments **quarks** and **gluons** observed as jets of particles



QCD and Colour in e^+e^- Collisions

★ e^+e^- colliders are an excellent place to study QCD

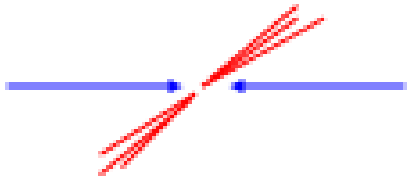


★ Well defined production of quarks
- QED process well-understood
- no need to know parton structure functions
- + experimentally very clean – no proton remnants

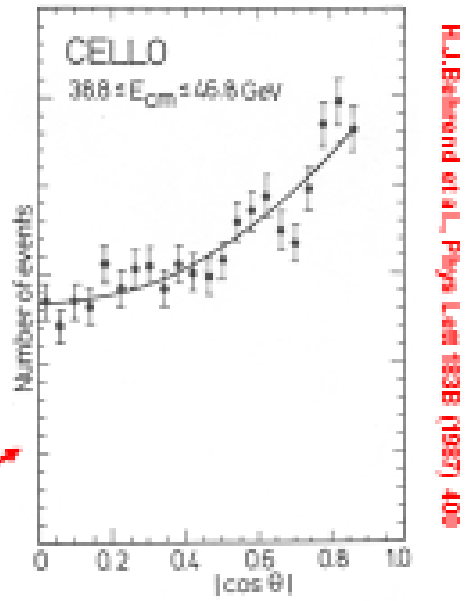
★ In handout 5 obtained expressions for the $e^+e^- \to \mu^+\mu^-$ cross-section

$$\sigma = \frac{4\pi\alpha^2}{3s} \qquad \frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4s}(1+\cos^2\theta)$$

- In e^+e^- collisions produce all quark flavours for which $\sqrt{s} > 2m_q$
- In general, i.e. unless producing a $q\bar{q}$ bound state, produce jets of hadrons
- Usually can't tell which jet came from the quark and came from anti-quark



★ Angular distribution of jets $\propto (1+\cos^2\theta)$

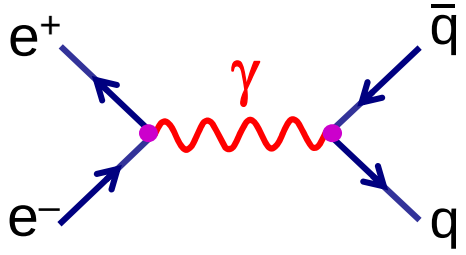


H.J. Behrend et al., Phys. Lett. 183B (1987) 400

➡ **Quarks are spin ½**

QCD and Colour in e^+e^- Collisions

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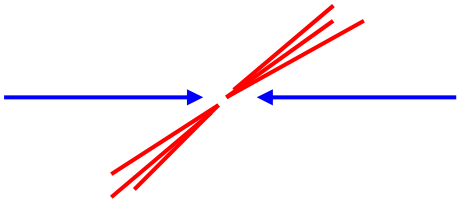


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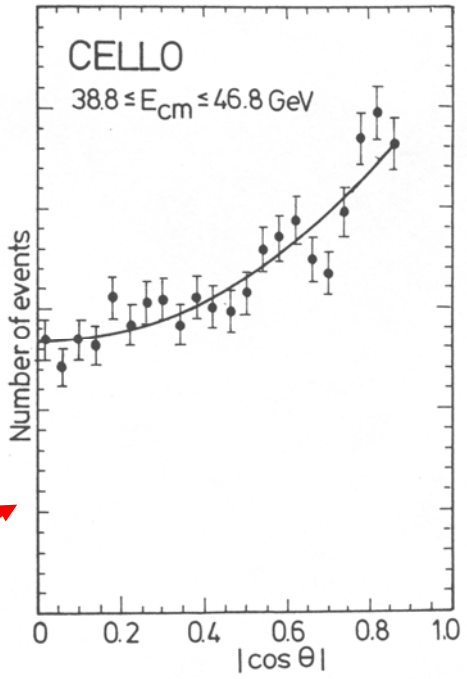
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- In e^+e^- collisions produce all quark flavours for which $\sqrt{s} > 2m_q$
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- Usually can't tell which jet came from the quark and came from anti-quark



★ Angular distribution of jets $\propto (1+\cos^2\theta)$



H.J.Behrend et al., Phys Lett 183B (1987) 400

➡ **Quarks are spin ½**

★ **Colour is conserved and quarks are produced as** $r\bar{r},\ g\bar{g},\ b\bar{b}$

★ **For a single quark flavour and single colour**

$$\sigma(e^+e^- \to q_i\bar{q}_i) = \frac{4\pi\alpha^2}{3s}Q_q^2$$

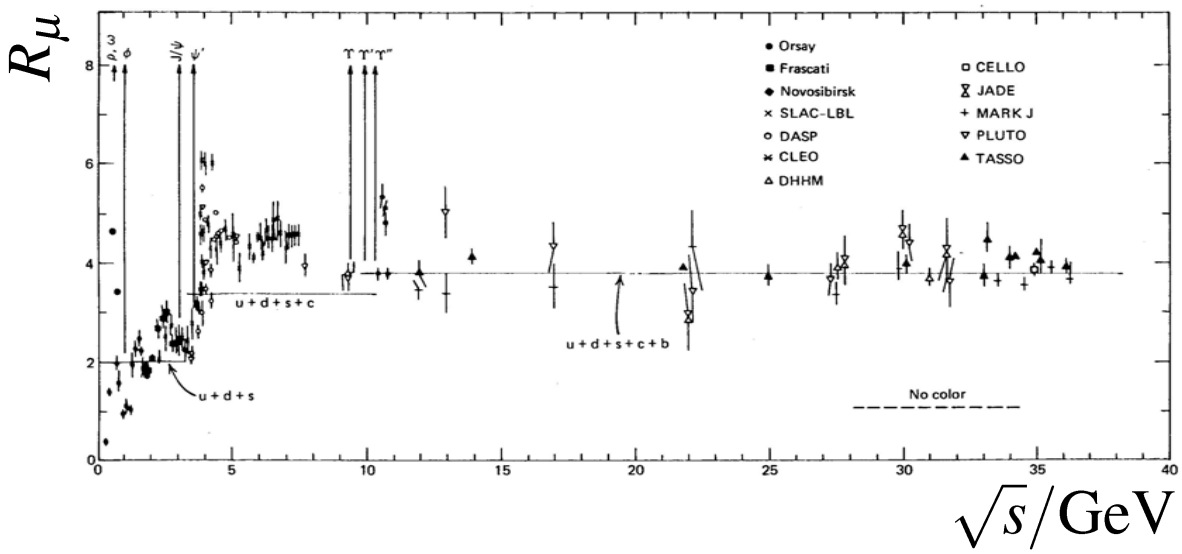
• **Experimentally observe jets of hadrons:**

$$\sigma(e^+e^- \to \text{hadrons}) = 3\sum_{u,d,s,..}\frac{4\pi\alpha^2}{3s}Q_q^2$$

Factor 3 comes from colours

• **Usual to express as ratio compared to** $\sigma(e^+e^- \to \mu^+\mu^-)$

$$R_\mu = \frac{\sigma(e^+e^- \to \text{hadrons})}{\sigma(e^+e^- \to \mu^+\mu^-)} = 3\sum_{u,d,s,..}Q_q^2$$



u,d,s: $R_\mu = 3\times(\frac{1}{9}+\frac{4}{9}+\frac{1}{9}) = 2$

u,d,s,c: $R_\mu = \frac{10}{3}$

u,d,s,c,b: $R_\mu = \frac{11}{3}$

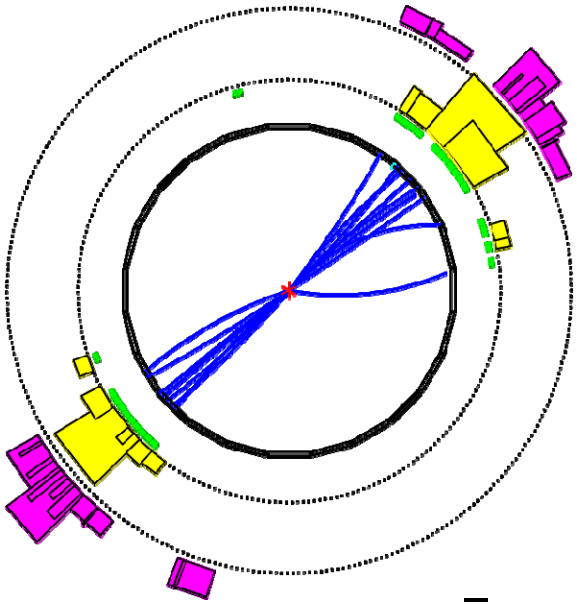
★**Data consistent with expectation with factor 3 from colour**

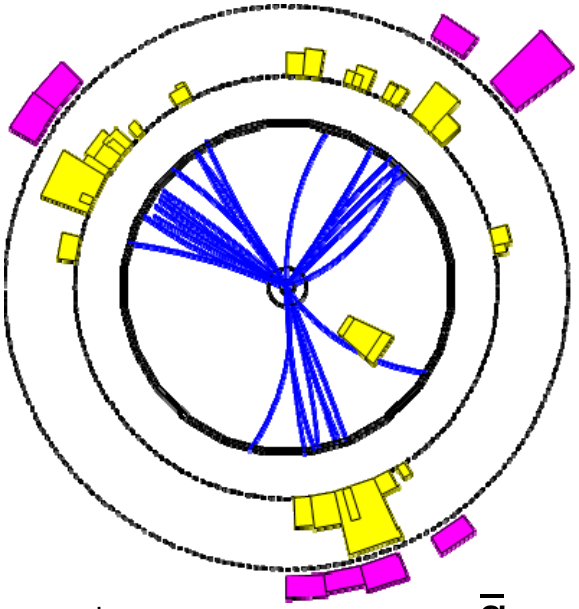
Jet production in e+e- Collisions

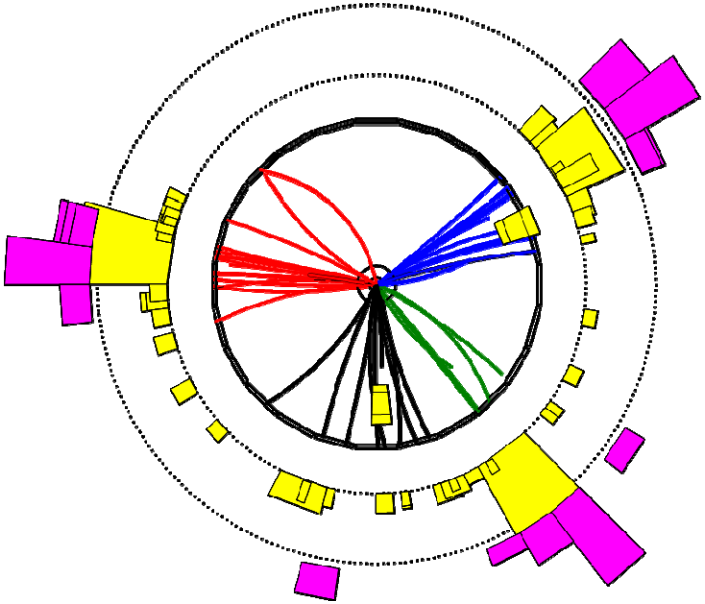
★e^+e^- colliders are also a good place to study gluons

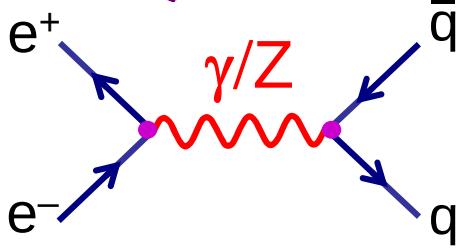
$e^+e^- \to q\bar{q} \to 2\text{jets}$ $e^+e^- \to q\bar{q}g \to 3\text{jets}$ $e^+e^- \to q\bar{q}gg \to 4\text{jets}$

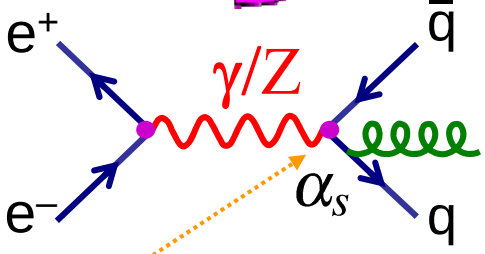
OPAL at LEP (1989-2000)

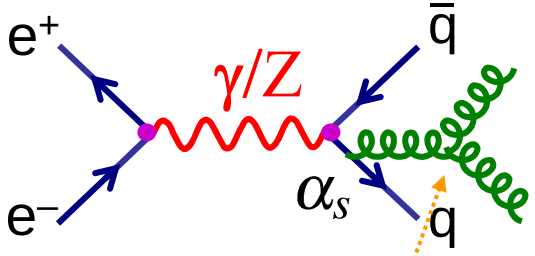












Experimentally:

- **Three jet rate ➡ measurement of α_s**
- **Angular distributions ➡ gluons are spin-1**
- **Four-jet rate and distributions ➡ QCD has an underlying SU(3) symmetry**

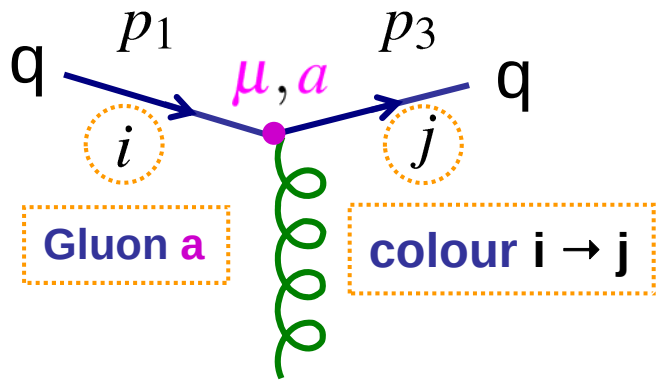
The Quark – Gluon Interaction

- **Representing the colour part of the fermion wave-functions by:**

$$r = c_1 = \begin{pmatrix}1\\0\\0\end{pmatrix} \qquad g = c_2 = \begin{pmatrix}0\\1\\0\end{pmatrix} \qquad b = c_3 = \begin{pmatrix}0\\0\\1\end{pmatrix}$$

- **Particle wave-functions** $u(p) \longrightarrow c_i u(p)$
- **The QCD** qqg **vertex is written:**

$$\bar{u}(p_3) c_j^\dagger \{-\tfrac{1}{2} i g_s \lambda^a \gamma^\mu\} c_i u(p_1)$$



- **Only difference w.r.t. QED is the insertion of the 3x3 SU(3) Gell-Mann matrices**
- **Isolating the colour part:**

$$c_j^\dagger \lambda^a c_i = c_j^\dagger \begin{pmatrix}\lambda^a_{1i}\\ \lambda^a_{2i}\\ \lambda^a_{3i}\end{pmatrix} = \lambda^a_{ji}$$

- **Hence the fundamental quark - gluon QCD interaction can be written**

$$\bar{u}(p_3) c_j^\dagger \{-\tfrac{1}{2} i g_s \lambda^a \gamma^\mu\} c_i u(p_1) \equiv \bar{u}(p_3) \{-\tfrac{1}{2} i g_s \lambda^a_{ji} \gamma^\mu\} u(p_1)$$

Feynman Rules for QCD

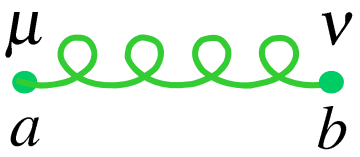
- **External Lines**

spin 1/2	incoming quark	$u(p)$	
	outgoing quark	$\bar{u}(p)$	
	incoming anti-quark	$\bar{v}(p)$	
	outgoing anti-quark	$v(p)$	
spin 1	incoming gluon	$\varepsilon^{\mu}(p)$	
	outgoing gluon	$\varepsilon^{\mu}(p)^*$	

- **Internal Lines (propagators)**

  spin 1 gluon

$$\frac{-ig_{\mu\nu}}{q^2}\delta^{ab}$$

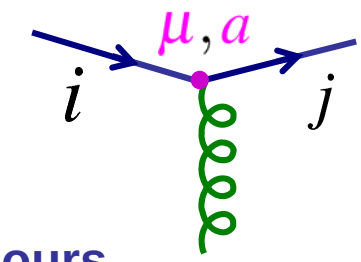


a, b = 1,2,…,8 are gluon colour indices

- **Vertex Factors**

  spin 1/2 quark

$$-ig_s\tfrac{1}{2}\lambda^a_{ji}\gamma^{\mu}$$



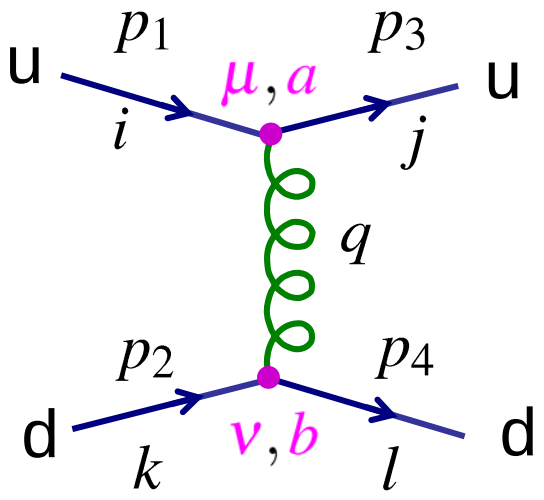
i, j = 1,2,3 are quark colours,

λ^a **a = 1,2,..8 are the Gell-Mann SU(3) matrices**

- **+ 3 gluon and 4 gluon interaction vertices**
- **Matrix Element** $-iM$ **= product of all factors**

Matrix Element for quark-quark scattering

★ **Consider QCD scattering of an up and a down quark**



- The incoming and out-going quark colours are labelled by $i, j, k, l = \{1,2,3\}$ (or $\{r, g, b\}$)
- In terms of colour this scattering is $ik \to jl$
- The 8 different gluons are accounted for by the colour indices $a, b = 1, 2, ..., 8$
- NOTE: the δ-function in the propagator ensures ***a*** = ***b***, i.e. the gluon "emitted" at ***a*** is the same as that "absorbed" at ***b***

★ **Applying the Feynman rules:**

$$-iM = \left[\bar{u}_u(p_3)\{-\tfrac{1}{2}ig_s\lambda^a_{ji}\gamma^\mu\}u_u(p_1)\right]\frac{-ig_{\mu\nu}}{q^2}\delta^{ab}\left[\bar{u}_d(p_4)\{-\tfrac{1}{2}ig_s\lambda^b_{lk}\gamma^\nu\}u_d(p_2)\right]$$

where summation over ***a*** and ***b*** (and μ and ν) is implied.

★ **Summing over *a* and *b* using the δ-function gives:**

$$M = -\frac{g_s^2}{4}\lambda^a_{ji}\lambda^a_{lk}\frac{1}{q^2}g_{\mu\nu}[\bar{u}_u(p_3)\gamma^\mu u_u(p_1)][\bar{u}_d(p_4)\gamma^\nu u_d(p_2)]$$

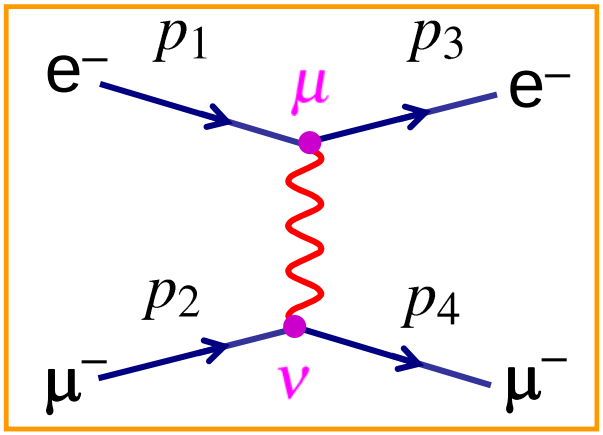
Sum over all 8 gluons (repeated indices)

QCD vs QED

QED

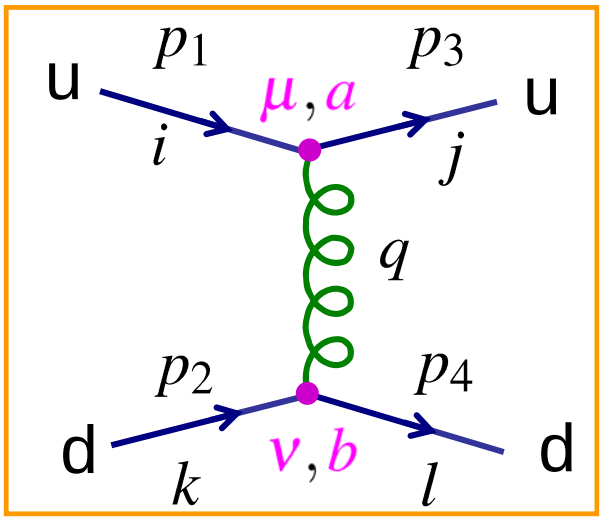
$$-iM = [\bar{u}(p_3) ie\gamma^\mu u(p_1)] \frac{-ig_{\mu\nu}}{q^2} [\bar{u}(p_4) ie\gamma^\nu u(p_2)]$$

$$M = -e^2 \frac{1}{q^2} g_{\mu\nu} [\bar{u}(p_3)\gamma^\mu u(p_1)][\bar{u}(p_4)\gamma^\nu u(p_2)]$$



QCD

$$M = -\frac{g_s^2}{4} \lambda^a_{ji} \lambda^a_{lk} \frac{1}{q^2} g_{\mu\nu} [\bar{u}_u(p_3)\gamma^\mu u_u(p_1)][\bar{u}_d(p_4)\gamma^\nu u_d(p_2)]$$



★ **QCD Matrix Element = QED Matrix Element with:**

- $e^2 \to g_s^2$ **or equivalently** $\alpha = \frac{e^2}{4\pi} \to \alpha_s = \frac{g_s^2}{4\pi}$

+ QCD Matrix Element includes an additional "colour factor"

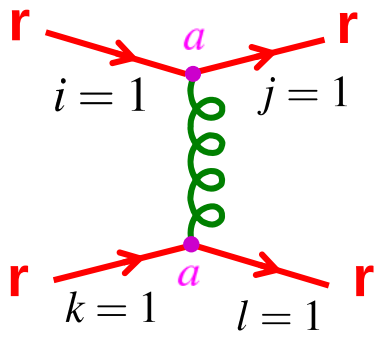
$$C(ik \to jl) \equiv \frac{1}{4} \sum_{a=1}^{8} \lambda^a_{ji} \lambda^a_{lk}$$

Evaluation of QCD Colour Factors

- **QCD colour factors reflect the gluon states that are involved**

$\lambda^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$	$\lambda^4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$	$\lambda^6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$	$\lambda^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
$\lambda^2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$	$\lambda^5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}$	$\lambda^7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}$	$\lambda^8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$
Gluons: $r\bar{g}, g\bar{r}$	$r\bar{b}, b\bar{r}$	$g\bar{b}, b\bar{g}$	$\frac{1}{\sqrt{2}}(r\bar{r} - g\bar{g})$ $\frac{1}{\sqrt{6}}(r\bar{r} + g\bar{g} - 2b\bar{b})$

❶ Configurations involving a single colour



- **Only matrices with non-zero entries in 11 position are involved**

$$C(rr \to rr) = \frac{1}{4}\sum_{a=1}^{8} \lambda^a_{11}\lambda^a_{11} = \frac{1}{4}(\lambda^3_{11}\lambda^3_{11} + \lambda^8_{11}\lambda^8_{11})$$

$$= \frac{1}{4}\left(1 + \frac{1}{3}\right) = \frac{1}{3}$$

Similarly find $C(rr \to rr) = C(gg \to gg) = C(bb \to bb) = \frac{1}{3}$

Color Factors

Halzen&Martin :

Most simple QCD diagrams are exactly analogous to QED diagrams, and the QCD cross section is obtained by the replacement

$$\alpha^n \to C_F \alpha_s^n, \tag{10.18}$$

Quark and Quark

We turn now to the interaction of two quarks. Again, we shall assume that they have different flavors, so the only diagram (in lowest order) is the one indicated in Figure (8.7),* representing, say, $u + d \to u + d$. The amplitude is

$$\mathcal{M} = \frac{-g_s^2}{4}\frac{1}{q^2}[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)](c_3^\dagger \lambda^\alpha c_1)(c_4^\dagger \lambda^\alpha c_2) \tag{8.56}$$

This part: totally the same as QED right part cLc, completely factorizes like photon epsilon vectors, into color factor --> just norm factor difference, angular dependence e.g. is the same.